

Orientation-Dependent Dispersion, Directional Band Gaps, and Wave Steering in Two-Dimensional Strain-Gradient Elasticity

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Abstract

Periodic microstructured solids can show direction-dependent dispersion when the wavelength approaches an internal material length. We combine a rotated ellipsoidal characteristic-length tensor with local Mindlin Form-II strain-gradient elasticity and independent gradient micro-inertia. The two-dimensional Bloch–Floquet problem is discretized with a conforming 32-degree-of-freedom Bogner–Fox–Schmit element; the same complex Bloch phase is applied to displacement and derivative degrees of freedom. In the homogeneous case, orientation and aspect ratio shift the branches and produce directional stop bands. No complete gap is found ($\Delta_{\text{complete}} \leq -0.3235$). The largest sampled directional gap is $\Delta_{GX} = 0.0677$ at $\theta = 0^\circ$, $\text{AR} = 7$, and the orientation sensitivity reaches $S_\theta = 2.34 \text{ rad}^{-1}$ at $\text{AR} = 10$. On the acoustic branch at $\bar{k} = 0.5$, the maximum group-velocity deviation is 2.78° for $\text{AR} = 10$. Across the sampled orientations, the steering lobes move while their peak deviation remains invariant within 10^{-5} deg. A separate high-contrast phononic-crystal case has a positive complete gap of 2.5732 (43.94%) on the 4×4 mesh. The gap falls to 1.9179 at 64×64 and is not mesh-converged. Gradient micro-inertia bounds the high-wavenumber phase speed for the branch studied. The results distinguish orientation-based directional control in a homogeneous medium from complete attenuation enabled by material contrast.

Keywords: Strain-gradient elasticity, Mindlin Form-II, Micro-inertia, Bloch–Floquet analysis, C^1 Bogner–Fox–Schmit element, Anisotropic length tensor, Directional band gaps, Wave steering

1. Introduction

Periodic elastic media can be designed to control wave propagation through their dispersion. Early calculations established band structures for periodic composites; experiments and theory later showed that two-dimensional solids can support absolute acoustic gaps [1, 2, 3]. Lattice analyses and topology-optimization studies have since used geometry to tailor pass bands, stop bands, and wave transmission [4, 5, 6]. Recent methods target gaps at specified frequencies and use multiple materials to control gap formation, while scaling studies track how optimized gaps shift with structural size [7, 8, 9]. Reviews cover phononic-crystal mechanisms, engineered band gaps, and applications [10, 11, 12]. A gap along selected directions does not guarantee a complete gap; the latter requires suppression throughout the Brillouin zone.

When a wavelength approaches the material’s microstructural scale, a local Cauchy law may not capture size-dependent stiffness or dispersion. Higher-order theories add kinematic measures and their work-conjugate stresses. Their development draws on couple-stress and microstructure theories, second-gradient elasticity, and variational treatments of higher-order boundary work [13,

14, 15, 16, 17, 18]. Later studies examined strain-gradient descriptions of nonuniform deformation and small-scale beam experiments [19, 20, 21]. Reviews and lattice-based derivations describe other formulations of gradient elasticity [22, 23]. We use the local Mindlin Form-II theory; nonlocal integral and micromorphic theories are related approaches, but they are distinct formulations.

A dynamic gradient model needs a kinetic-energy law as well as gradient stiffness. With classical inertia alone, the shear branch examined here has phase speed proportional to wavenumber at high wavenumbers. Independent gradient micro-inertia gives this branch a finite high-wavenumber limit. That result follows from the constitutive and kinetic-energy laws used here; other higher-order continua need not have the same limit. One-dimensional studies have examined dispersion in gradient-elastic beams and periodic media, including dipolar-gradient formulations [24, 25, 26]. A two-dimensional phononic crystal has also been analyzed with a relaxed micromorphic model that includes free and gradient micro-inertia [27]. Together, these studies show that the continuum and inertia laws affect the interpretation of short-wavelength bands, although their constitutive models differ from the local Mindlin Form-II model used here.

Size-dependent phononic-crystal studies use a range of constitutive models. Transfer-matrix analyses show

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that imperfect interfaces can alter one-dimensional band gaps [28]. Other one-dimensional work considers dipolar-gradient elasticity, gradient thermoelasticity, and flexoelectric strain-gradient coupling [25, 29, 30, 26]. In two dimensions, researchers have applied nonlocal strain-gradient theory to nanoscale phononic crystals and studied mixed and defect modes in a phononic-crystal slab [31, 32]. Other model-specific comparisons include thermoelastic cylindrical crystals and functionally graded lattices [33, 34]. Anisotropy and orientation have also been studied in conventional phononic crystals: work on orthotropic fillers and anisotropic plates reports shifts in band locations and widths as material axes or inclusions rotate relative to the lattice [35, 36]. Recent studies examine direction-dependent gaps in anisotropic nanocrystal assemblies and the effects of anisotropic constituents in optimized phononic crystals [37, 38]. For flexural plate waves, anisotropic equal-frequency contours can align group velocity with a lens interface over a broad frequency range [39]. Prior work has therefore established orientation-dependent band tuning. Here we examine how an oriented characteristic-length tensor affects dispersion and energy-flow direction in a local two-dimensional strain-gradient model with gradient micro-inertia.

The finite-element approximation must also accommodate the strain-gradient weak form. Because it contains second derivatives of displacement, the trial space must be H^2 -conforming or use an equivalent mixed formulation. The bicubic Bogner–Fox–Schmit (BFS) rectangle provides C^1 continuity across elements on structured quadrilateral meshes [40]. Earlier studies developed displacement-based, mixed, and implicit-gradient finite elements [41, 42, 43, 44]. Later work treats second-gradient boundary-value problems, higher-order Hermite mappings on regular and distorted domains, mixed elements with patch recovery or shared degrees of freedom, and mixed crack formulations [45, 46, 47, 48, 49, 50]. For a conforming BFS Bloch element, the same phase transformation must be applied to displacement values and all derivative degrees of freedom, including the mixed derivative.

The studies cited above do not combine, in one two-dimensional Bloch–Floquet finite-element framework, a rotated ellipsoidal second-moment characteristic-length tensor, local Mindlin Form-II strain-gradient elasticity with independent gradient micro-inertia, and a conforming C^1 BFS discretization with phase transformations on all relevant degrees of freedom. This study addresses that combination and builds on established work on anisotropy, orientation effects, and size-dependent phononic crystals. It makes four contributions:

- We formulate a Bloch–Floquet framework for two-dimensional anisotropic strain-gradient elasticity. The rotated ellipsoidal averaging domain defines the length tensor, with aspect ratio $\text{AR} = l_1/l_2$ and orientation θ as separate design variables.
- Gradient micro-inertia is examined analytically and

numerically. For the branch studied, $\ell_i = 0$ gives an unbounded high-wavenumber phase speed, whereas $\ell_i > 0$ yields the finite limit $v_{T,\infty} = \sqrt{\mu/\rho} l_{\text{eff}}/(\sqrt{10} \ell_i)$ derived in Appendix A.

- A conforming 32-degree-of-freedom BFS element is developed, with the same complex Bloch phase applied to displacement, first derivatives, and the mixed derivative.
- Across the (θ, AR) design space, we evaluate directional dispersion, gap measures, and group velocity. The energy-flux relation follows from local balance; the analysis also separates directional stop bands from complete gaps and measures steering amplitude and lobe orientation.

Section 2 defines the anisotropic strain-gradient continuum and micro-inertia. Section 3 sets out the Bloch–Floquet formulation, followed by the BFS discretization in Section 4. Section 5 reports the analytical, numerical, and mesh checks. Section 6 presents the homogeneous dispersion, directional gaps, design maps, steering, and the separate heterogeneous Case C result; Section 7 examines the high-wavenumber response. Sections 8 and 9 interpret the findings and state their limitations.

2. Anisotropic Strain-Gradient Continuum with Micro-Inertia

2.1. Ellipsoidal characteristic-length tensor

In standard isotropic gradient elasticity, the microstructural influence is often represented by a scalar length scale l . Under plane strain, we describe an oriented microstructure by its in-plane elliptical averaging neighborhood $\mathcal{E}_0$, with principal semi-axes l_1 and l_2 :

$$\mathcal{E}_0 = \left\{ (x'_1, x'_2) \in \mathbb{R}^2 \mid \frac{(x'_1)^2}{l_1^2} + \frac{(x'_2)^2}{l_2^2} \leq 1 \right\}, \quad (1)$$

where (x'_1, x'_2) are local principal coordinates aligned with the microstructural axes. We define the in-plane characteristic-length tensor as the diagonal matrix of squared semi-axes. This model definition differs from the normalized covariance tensor of a uniformly filled two-dimensional ellipse:

$$\mathbf{L}_0 = \begin{bmatrix} l_1^2 & 0 \\ 0 & l_2^2 \end{bmatrix}. \quad (2)$$

We hold the area $\pi l_1 l_2$ fixed as the aspect ratio varies, using an isotropic baseline $l_{\text{iso}} > 0$:

$$l_1 = l_{\text{iso}} \sqrt{\text{AR}}, \quad l_2 = \frac{l_{\text{iso}}}{\sqrt{\text{AR}}}, \quad \pi l_1 l_2 = \pi l_{\text{iso}}^2, \quad \det(\mathbf{L}_0) = l_1^2 l_2^2 = l_{\text{iso}}^4 \quad (3)$$

where $\text{AR} = l_1/l_2 \geq 1$ is the microstructural aspect ratio. The fixed factor $1/10$ in the constitutive law below is the three-dimensional second-moment normalization adopted by the model; only the in-plane length components enter the plane-strain tensor.

Orientation convention and passive rotation. We take positive θ to denote a clockwise orientation of the principal axes relative to the global frame (x_1, x_2) . The passive map from global coordinates $\mathbf{x}$ to local principal coordinates $\mathbf{x}'$ is:

$$\mathbf{x}' = \mathbf{R}(\theta)\mathbf{x}, \quad \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (4)$$

Under this coordinate transformation, the characteristic-length tensor in the global frame is:

$$\mathbf{L}(\theta) = \mathbf{R}(\theta)^\top \mathbf{L}_0 \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} l_1^2 & 0 \\ 0 & l_2^2 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (5)$$

Carrying out the matrix multiplication gives:

$$L_{11}(\theta) = l_1^2 \cos^2 \theta + l_2^2 \sin^2 \theta, \quad (6a)$$

$$L_{22}(\theta) = l_1^2 \sin^2 \theta + l_2^2 \cos^2 \theta, \quad (6b)$$

$$L_{12}(\theta) = L_{21}(\theta) = (l_2^2 - l_1^2) \cos \theta \sin \theta. \quad (6c)$$

For an elongated microstructure ($l_1 > l_2$), the off-diagonal coupling $L_{12}(\theta)$ is negative for $0 < \theta < 90^\circ$ and reaches its minimum, $(l_2^2 - l_1^2)/2$, at $\theta = 45^\circ$. Because $\mathbf{R}(\theta)$ is orthogonal, the eigenvalues of $\mathbf{L}(\theta)$ are invariant under rotation:

$$\text{spec}(\mathbf{L}(\theta)) \equiv \{l_1^2, l_2^2\} \quad \forall \theta \in [0, 2\pi). \quad (7)$$

For $l_1, l_2 > 0$, $\mathbf{L}(\theta)$ is positive definite at every orientation.

2.2. Kinematics and constitutive response

Assuming infinitesimal deformations under two-dimensional plane-strain conditions ($u_3 = 0, \partial(\cdot)/\partial x_3 = 0$), the classical displacement gradient and infinitesimal strain tensor $\boldsymbol{\varepsilon}$ are:

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}), \quad i, j \in \{1, 2\}. \quad (8)$$

The strain-gradient tensor $\boldsymbol{\eta}$ is the spatial derivative of infinitesimal strain:

$$\eta_{ijk} = \varepsilon_{ij,k} = \frac{1}{2}(u_{i,jk} + u_{j,ik}), \quad \eta_{ijk} = \eta_{jik}. \quad (9)$$

Under plane strain in 2D, $\boldsymbol{\varepsilon}$ has 3 independent components ($\varepsilon_{11}, \varepsilon_{22}, 2\varepsilon_{12}$), and $\boldsymbol{\eta}$ has 6 ($\eta_{111}, \eta_{112}, \eta_{221}, \eta_{222}, 2\eta_{121}, 2\eta_{122}$).

Mindlin Form-II constitutive law. In Mindlin's Form II, the strain-energy density W has classical and strain-gradient contributions:

$$W = \frac{1}{2}\sigma_{ij}\varepsilon_{ij} + \frac{1}{2}\tau_{ijk}\eta_{ijk}, \quad (10)$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\boldsymbol{\tau}$ is the third-order double-stress (or dipolar stress) tensor. For an isotropic base material parameterized by Lamé constants λ and μ , the Cauchy stress satisfies Hooke's law:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} = \lambda\delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}. \quad (11)$$

The double-stress tensor τ_{ijk} is conjugate to η_{ijk} through the anisotropic gradient-elasticity law involving $\mathbf{L}$:

$$\tau_{ijk} = \frac{1}{10}L_{kl}C_{ijmn}\eta_{mnl} = \frac{1}{10}L_{kl}(\lambda\delta_{ij}\eta_{mml} + 2\mu\eta_{ijl}), \quad (12)$$

For stable isotropic elastic constants ($\mu > 0$ and $3\lambda + 2\mu > 0$) and positive-definite $\mathbf{L}$, the quadratic strain energy is non-negative and vanishes only when both the strain and strain-gradient fields vanish.

Gradient micro-inertia. We include gradient micro-inertia in the kinetic-energy density T :

$$T = \frac{1}{2}\rho\dot{u}_i\dot{u}_i + \frac{1}{2}\rho\ell_i^2\dot{u}_{i,j}\dot{u}_{i,j}, \quad (13)$$

where ρ is the macroscopic mass density and $\ell_i > 0$ is the independent micro-inertia length parameter. Applying Hamilton's principle $\delta \int_{t_1}^{t_2} (T - W) dt = 0$ yields the strong-form equations of dynamic equilibrium in the bulk:

$$\sigma_{ij,j} - \tau_{ijk,jk} = \rho(\ddot{u}_i - \ell_i^2\ddot{u}_{i,jj}), \quad \text{in } \Omega. \quad (14)$$

The operator $1 - \ell_i^2\nabla^2$ increases the effective inertia of spatially varying acceleration. For the shear branch examined here, Section 7 and Appendix A show that this term bounds the high-wavenumber phase speed.

Model reductions. The model has four useful physical limits:

- (i) **Classical elasticity limit** ($l_{\text{iso}} \rightarrow 0$ at fixed finite AR, $\ell_i \rightarrow 0$): Both principal gradient lengths vanish, double stresses vanish, and the classical Navier equations $\sigma_{ij,j} = \rho\ddot{u}_i$ are recovered with dispersionless acoustic speeds $c_T = \sqrt{\mu/\rho}$ and $c_L = \sqrt{(\lambda + 2\mu)/\rho}$.
- (ii) **Isotropic gradient elasticity** (AR = 1): Here $l_1 = l_2 = l_{\text{iso}}$, so $L_{11} = L_{22} = l_{\text{iso}}^2$ and $L_{12} = 0$ for all θ , recovering isotropic Form-II gradient elasticity.
- (iii) **Gradient elasticity without micro-inertia** ($\ell_i = 0$): Higher-order stiffness remains while dynamic regularization is removed; the transverse branch analyzed in Appendix A then has $v_p \propto k$ as $k \rightarrow \infty$.
- (iv) **Quarter-turn covariance:** For fixed l_1, l_2 , a 90° change of orientation interchanges the diagonal components and reverses the off-diagonal component: $L_{11}(\theta + 90^\circ) = L_{22}(\theta)$, $L_{22}(\theta + 90^\circ) = L_{11}(\theta)$, and $L_{12}(\theta + 90^\circ) = -L_{12}(\theta)$.

2.3. Boundary operators

Let $\Gamma = \partial\Omega$ be a smooth boundary with unit outward normal $\mathbf{n}$. For the strain-energy term, integration by parts gives the boundary contribution

$$\delta \int_{\Omega} W d\Omega \Big|_{\Gamma} = \int_{\Gamma} [(\sigma_{ij} - \tau_{ijk,k})n_j \delta u_i + \tau_{ijk}n_k \delta u_{i,j}] d\Gamma. \quad (15)$$

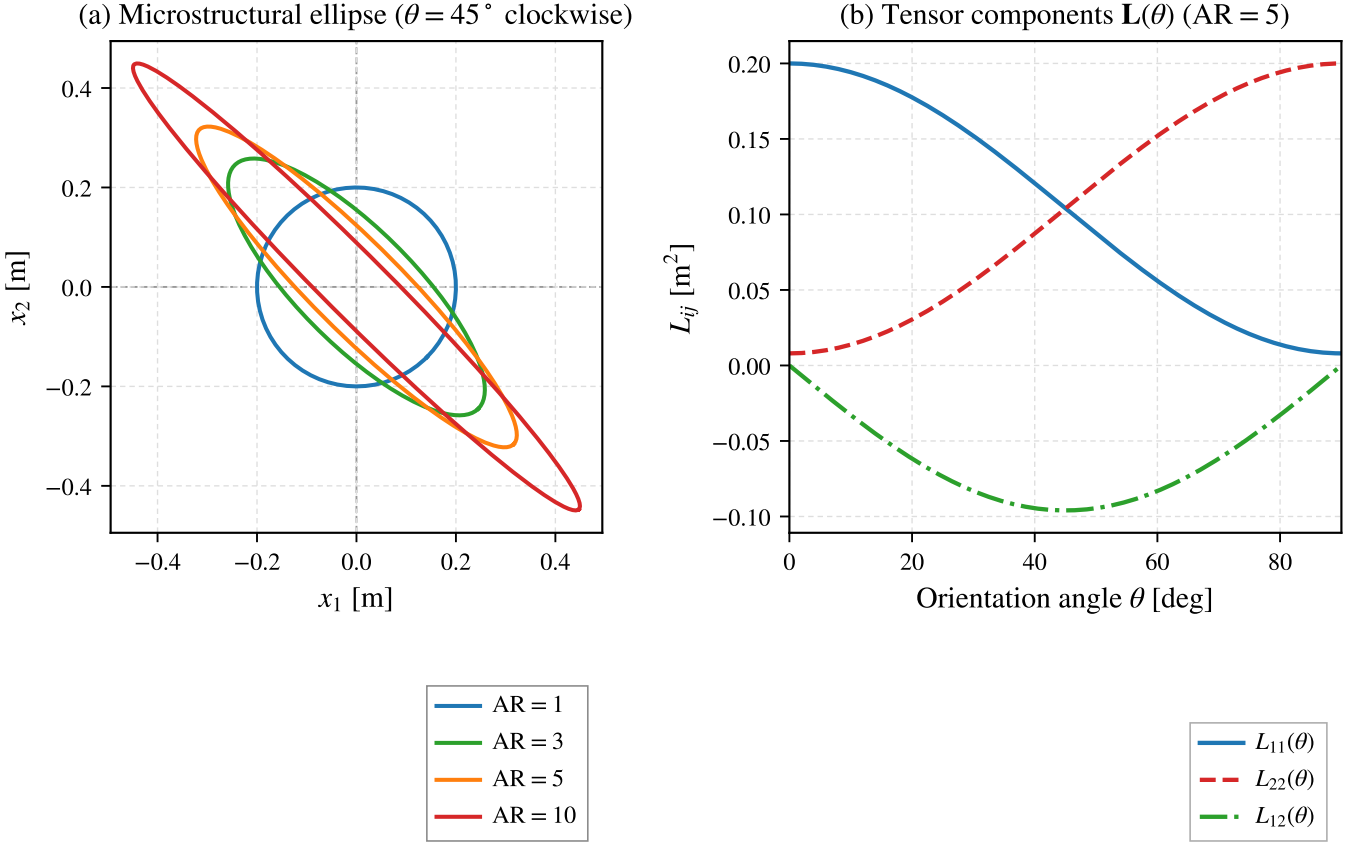


Figure 1: (a) Area-preserving averaging ellipses at $\theta = 45^\circ$ for AR = 1, 3, 5, 10, with positive orientation measured clockwise. (b) Length-tensor components versus orientation for AR = 5, including $L_{12}(45^\circ) = -0.096 \text{ m}^2$ under the same passive-rotation convention.

On a smooth boundary, decomposing $\delta u_{i,j}$ into normal and tangential parts and integrating the tangential part by parts gives the work-conjugate double traction and effective elastic traction:

$$R_i = n_j n_k \tau_{ijk}, \quad p_i^{\text{el}} = (\sigma_{ij} - \tau_{ijk,k}) n_j - \nabla_j^S (n_k \tau_{ijk}), \quad (16)$$

where ∇_j^S denotes the surface divergence acting on the tangential projection of $n_k \tau_{ijk}$; additional line forces arise at boundary edges. The gradient micro-inertia adds $\rho \ell_i^2 \ddot{u}_{i,j} n_j$ to the dynamic traction, so $p_i^{\text{dyn}} = p_i^{\text{el}} + \rho \ell_i^2 \ddot{u}_{i,j} n_j$. The work-conjugate pairs are (u_i, p_i^{dyn}) and $(\partial u_i / \partial n, R_i)$. On Bloch-paired cell faces, the kinematic phase constraints replace independent free-boundary conditions.

3. Bloch–Floquet formulation and observables

3.1. Square lattice and Bloch conditions

We generate the two-dimensional periodic medium by translating the square cell $\Omega_{\text{cell}} = [0, L] \times [0, L]$ along

$$\mathbf{a}_1 = \begin{bmatrix} L \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ L \end{bmatrix}. \quad (17)$$

The corresponding reciprocal vectors satisfy $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}$:

$$\mathbf{b}_1 = \frac{2\pi}{L} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \frac{2\pi}{L} \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (18)$$

The first Brillouin zone is $\mathcal{B} = [-\pi/L, \pi/L]^2$. For the isotropic square lattice, C_{4v} symmetry permits the path $\Gamma(0,0) \rightarrow X(\pi/L, 0) \rightarrow M(\pi/L, \pi/L) \rightarrow \Gamma$. Anisotropy reduces the symmetry to C_{2v} for $\theta = 0^\circ$ or 90° , and to C_2 at general orientation; full-zone measures are therefore evaluated over the corresponding Brillouin zone.

For a time-harmonic displacement $\mathbf{u}(\mathbf{x}, t) = \tilde{\mathbf{u}}(\mathbf{x}) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, Bloch periodicity gives

$$\mathbf{u}(\mathbf{x} + \mathbf{a}_\alpha) = \mathbf{u}(\mathbf{x}) e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}, \quad \alpha \in \{1, 2\}. \quad (19)$$

Because the approximation is C^1 -conforming, differentiating the displacement relation gives:

$$\frac{\partial \mathbf{u}}{\partial x_\beta}(\mathbf{x} + \mathbf{a}_\alpha) = \frac{\partial \mathbf{u}}{\partial x_\beta}(\mathbf{x}) e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}, \quad \frac{\partial^2 \mathbf{u}}{\partial x_1 \partial x_2}(\mathbf{x} + \mathbf{a}_\alpha) = \frac{\partial^2 \mathbf{u}}{\partial x_1 \partial x_2}(\mathbf{x}) e^{i\mathbf{k} \cdot \mathbf{a}_\alpha} \quad (20)$$

Thus displacement, first derivatives, and cross derivatives all take the same Bloch phase; a different phase for derivative degrees of freedom would break C^1 continuity across cell boundaries.

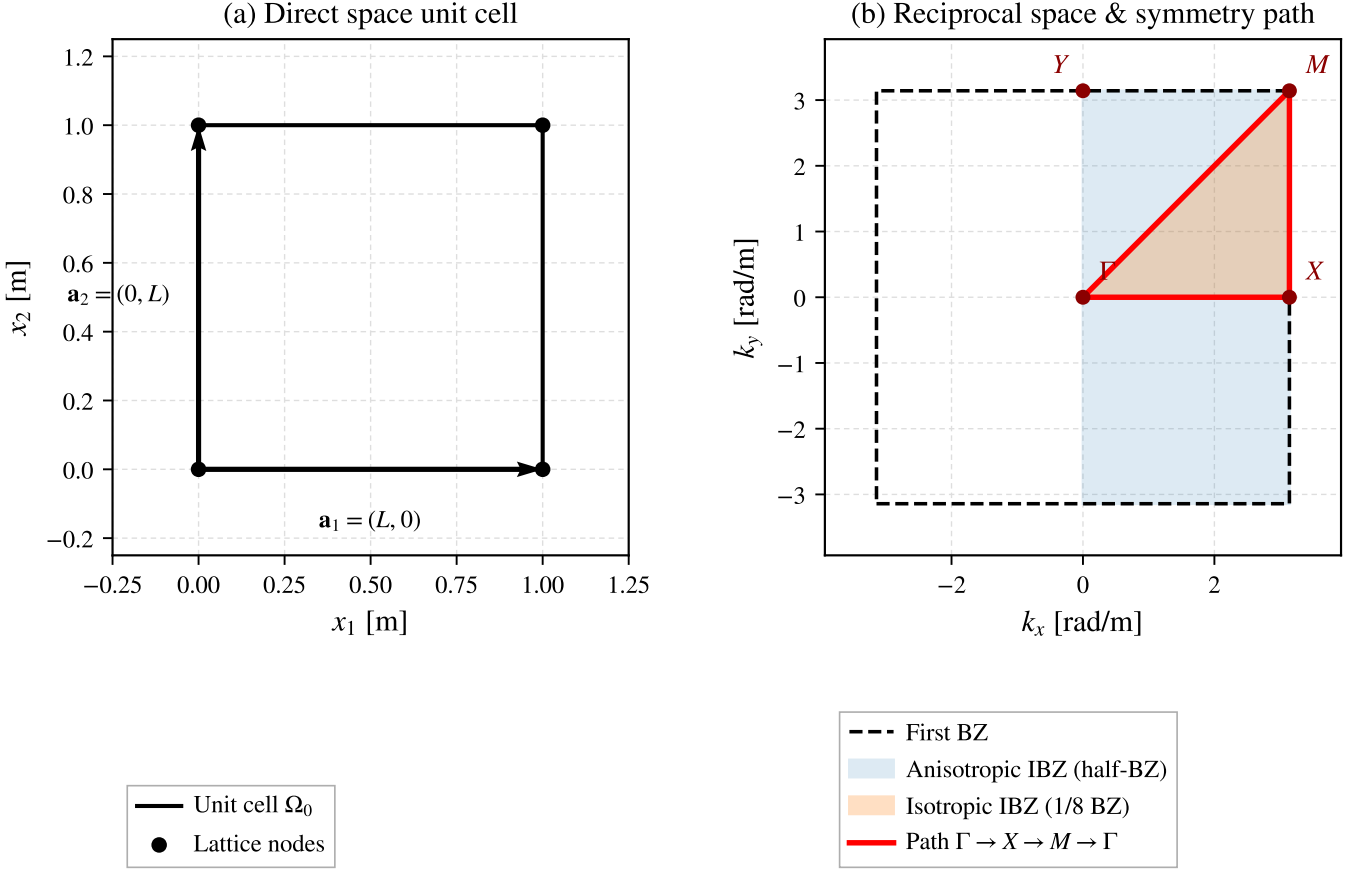


Figure 2: (a) Square periodic lattice. (b) First Brillouin zone and the high-symmetry contour Γ -X-M- Γ .

3.2. Nondimensionalization

For Case H, we scale length, frequency, and velocity with the cell length L , shear modulus μ , and density ρ :

$$\bar{k} = \frac{kL}{\pi}, \quad \bar{\omega} = \frac{\omega}{\omega_0}, \quad \omega_0 = \sqrt{\frac{\mu}{\rho L^2}}, \quad \bar{l}_m = \frac{l_m}{L}, \quad \bar{\ell}_i = \frac{\ell_i}{L} \quad (21)$$

This gives $c_T = 1$ and $c_L = \sqrt{(\lambda + 2\mu)/\mu} = \sqrt{3}$ for $\lambda/\mu = 1$ ($\nu = 0.25$). Case C uses the matrix shear speed and frequency scale specified in Table 2.

3.3. Gap and steering measures

At each wave vector, the eigenfrequencies are ordered as $\bar{\omega}_1(\mathbf{k}) \leq \bar{\omega}_2(\mathbf{k}) \leq \dots$. We distinguish gaps evaluated on one path leg, the complete high-symmetry path, and the full Brillouin zone:

(a) **Leg gap:** for bands n and $n + 1$, $\Delta[\text{leg}] = \min_{\mathbf{k} \in \text{leg}} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \text{leg}} \bar{\omega}_n(\mathbf{k})$; Δ_{GX} denotes the Γ -X leg.

(b) **Path gap:**

$$\Delta_{\text{path}} = \min_{\mathbf{k} \in \text{path}} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \text{path}} \bar{\omega}_n(\mathbf{k}). \quad (22)$$

(c) **Complete gap:**

$$\Delta_{\text{complete}} = \min_{\mathbf{k} \in \mathcal{B}} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \mathcal{B}} \bar{\omega}_n(\mathbf{k}). \quad (23)$$

Since the leg, path, and full zone are nested sets, $\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$.

$$\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]. \quad (24)$$

A positive leg gap indicates directional filtering; a complete band gap requires $\Delta_{\text{complete}} > 0$.

The orientation sensitivity of a directional gap is measured by

$$S_\theta = \frac{\max_\theta \left| \frac{\partial \Delta}{\partial \theta} \right|}{\max_\theta \Delta} \quad [\text{rad}^{-1}]. \quad (25)$$

The group velocity and its angular deviation from the wave vector are

$$\mathbf{v}_g = \nabla_{\mathbf{k}} \omega. \quad (26)$$

$$\delta = \angle \mathbf{v}_g - \angle \mathbf{k}. \quad (27)$$

4. C^1 Finite Element Discretisation

4.1. Bogner-Fox-Schmit Bicubic Hermite Element

The Mindlin Form-II weak form contains second derivatives of displacement, so its trial functions must lie in the

Sobolev space $H^2(\Omega)$. Standard C^0 Lagrange elements do not meet this conformity requirement. We use the Bogner–Fox–Schmit (BFS) bicubic element, which provides C^1 continuity across rectangular element boundaries on structured meshes [40].

For a rectangular element of dimensions $2a \times 2b$, we interpolate $u_1(x, y)$ and $u_2(x, y)$ independently, using tensor products of one-dimensional cubic Hermite polynomials:

$$u_i(x, y) = \sum_{a=1}^4 \sum_{p=1}^4 N_{a,p}(x, y) d_{i,a,p}^e, \quad i \in \{1, 2\}, \quad (28)$$

where $a \in \{1, 2, 3, 4\}$ indexes the four element vertices, and $p \in \{1, 2, 3, 4\}$ denotes the four nodal degrees of freedom per displacement component:

$$\mathbf{d}_{i,a}^e = \left[u_i \quad \frac{\partial u_i}{\partial x} \quad \frac{\partial u_i}{\partial y} \quad \frac{\partial^2 u_i}{\partial x \partial y} \right]_{\text{at node } a}^T. \quad (29)$$

With 4 vertices and 2 displacement components, each BFS element has $4 \times 2 \times 4 = 32$ degrees of freedom. Figure 3 shows the nodal DOFs and their Bloch-phase mapping at periodic boundaries. All reported band structures and design maps use a uniform mesh of $n \times n = 16 \times 16$ BFS elements per unit cell. This gives $h = L/n = 0.0625$ and $8n^2 = 2048$ degrees of freedom per cell; Table 2 lists the element order n .

4.2. Element matrices

The strain tensor $\boldsymbol{\varepsilon}$ and strain-gradient tensor $\boldsymbol{\eta}$ within element e are related to the element nodal displacement vector $\mathbf{d}^e \in \mathbb{R}^{32}$ via kinematic strain-displacement matrices $\mathbf{B}^c$ and $\mathbf{B}_k^g$:

$$\boldsymbol{\varepsilon} = \mathbf{B}^c \mathbf{d}^e, \quad \boldsymbol{\eta}_k \equiv \frac{\partial \boldsymbol{\varepsilon}}{\partial x_k} = \mathbf{B}_k^g \mathbf{d}^e, \quad k \in \{1, 2\}. \quad (30)$$

The element stiffness matrix $\mathbf{K}^e$ is the sum of classical and gradient contributions:

$$\mathbf{K}^e = \mathbf{K}^c + \mathbf{K}^g(\theta, \text{AR}), \quad (31)$$

where the classical stiffness matrix is:

$$\mathbf{K}^c = \int_{\Omega_e} (\mathbf{B}^c)^T \mathbf{C} \mathbf{B}^c \, d\Omega, \quad (32)$$

and the anisotropic gradient stiffness matrix is:

$$\mathbf{K}^g(\theta, \text{AR}) = \frac{1}{10} \sum_{k=1}^2 \sum_{l=1}^2 L_{kl}(\theta) \int_{\Omega_e} (\mathbf{B}_k^g)^T \mathbf{C} \mathbf{B}_l^g \, d\Omega. \quad (33)$$

Because the shape functions are bicubic tensor-product polynomials, each homogeneous-element stiffness and mass integrand has degree at most six in each coordinate. A four-point Gauss–Legendre rule is exact through degree seven, so the 4×4 tensor rule is the smallest rule that integrates all such terms exactly. This avoids quadrature error for homogeneous elements; the unconstrained stiffness still has its expected rigid-body null modes. In elements cut by the inclusion, the discontinuous material field is instead integrated by sub-cell Gauss sampling with an $O(h)$ geometric approximation (Section 6.7).

Mass and micro-inertia matrices. The kinetic energy gives the classical and micro-inertia mass contributions:

$$\mathbf{M}^e = \mathbf{M}_0 + \ell_1^2 \mathbf{M}^g, \quad (34)$$

with:

$$\mathbf{M}_0 = \rho \int_{\Omega_e} \mathbf{N}^T \mathbf{N} \, d\Omega, \quad \mathbf{M}^g = \rho \sum_{k=1}^2 \int_{\Omega_e} \left(\frac{\partial \mathbf{N}}{\partial x_k} \right)^T \left(\frac{\partial \mathbf{N}}{\partial x_k} \right) \, d\Omega. \quad (35)$$

Both $\mathbf{M}_0$ and $\mathbf{M}^g$ are symmetric and positive semi-definite; their sum $\mathbf{M}^e$ is strictly positive definite for any positive density $\rho > 0$.

4.3. Bloch reduction and eigenproblem

We partition the global degrees of freedom $\mathbf{d}$ into interior (I), master-boundary (M), and slave-boundary (S) sets to impose the Bloch conditions on $\partial\Omega_{\text{cell}}$:

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_I \\ \mathbf{d}_M \\ \mathbf{d}_S \end{bmatrix}. \quad (36)$$

The Bloch transformation relates the slave DOFs on the top and right boundaries to the corresponding master DOFs on the bottom and left boundaries:

$$\mathbf{d}_S = \mathbf{T}_{SM}(\mathbf{k}) \mathbf{d}_M, \quad (37)$$

where $\mathbf{T}_{SM}(\mathbf{k})$ is a diagonal complex phase matrix formed by evaluating $e^{ik_x L}$, $e^{ik_y L}$, or $e^{i(k_x + k_y)L}$ for each periodic node pair. As derived in Section 3, identical phase factors apply to the displacement, first-order, and cross-derivative DOFs.

In the continuous weak form, natural boundary work cancels between opposite cell faces. The effective traction p_i^{dyn} defined in Section 2 carries the Bloch phase and changes sign with the outward normal; its test trace carries the conjugate phase. The double traction $R_i = n_j n_k \tau_{ijk}$ also carries the Bloch phase but is unchanged when the normal reverses, while its work-conjugate normal derivative changes sign. These signs and phases cancel both face integrals. Line and corner terms pair under the same translations, including the product phase $e^{i(k_x + k_y)L}$ at the corner; no separate free-boundary condition is imposed on a Bloch-paired face.

We collect these relations in the master–slave transformation matrix $\mathbf{T}(\mathbf{k})$:

$$\mathbf{d} = \mathbf{T}(\mathbf{k}) \bar{\mathbf{d}}, \quad \mathbf{T}(\mathbf{k}) = \begin{bmatrix} \mathbf{I}_{II} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{MM} \\ \mathbf{0} & \mathbf{T}_{SM}(\mathbf{k}) \end{bmatrix}, \quad \bar{\mathbf{d}} = \begin{bmatrix} \mathbf{d}_I \\ \mathbf{d}_M \end{bmatrix}, \quad (38)$$

The reduced stiffness and mass matrices follow from Galerkin projection:

$$\bar{\mathbf{K}}(\mathbf{k}) = \mathbf{T}(\mathbf{k})^H \mathbf{K} \mathbf{T}(\mathbf{k}), \quad \bar{\mathbf{M}}(\mathbf{k}) = \mathbf{T}(\mathbf{k})^H \mathbf{M} \mathbf{T}(\mathbf{k}). \quad (39)$$

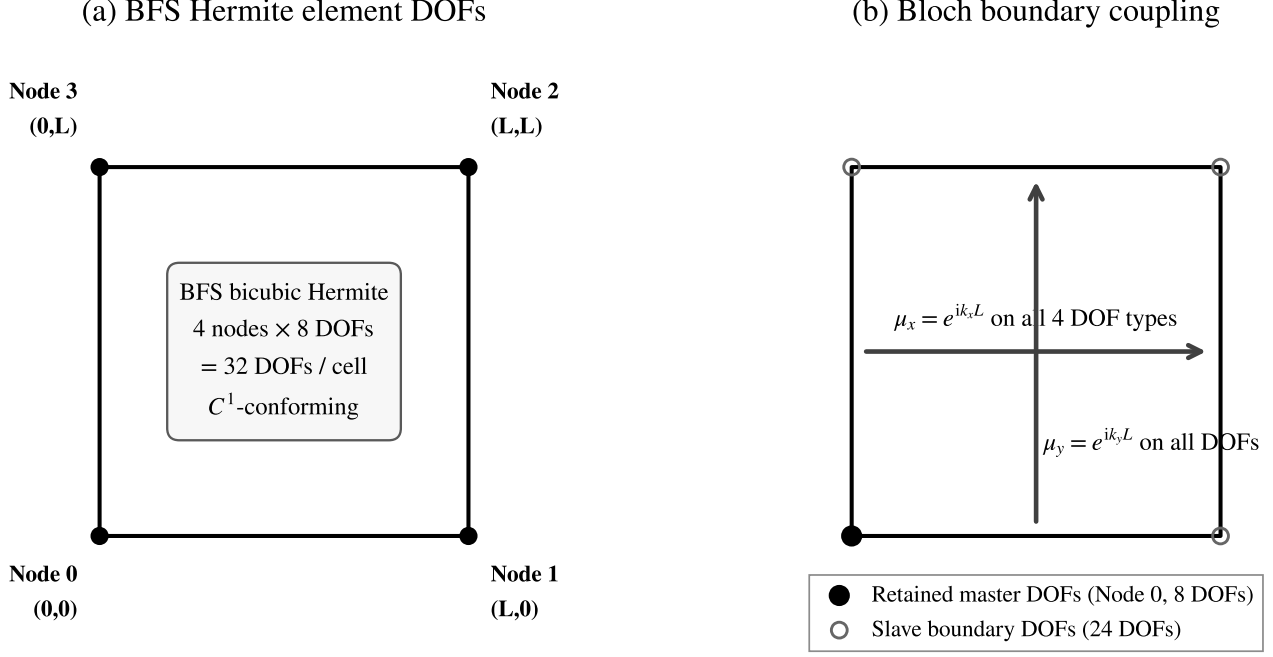


Figure 3: (a) Nodal degrees of freedom in the 32-DOF C^1 Bogner–Fox–Schmit (BFS) element; each displacement component has 4 nodal DOFs, $\{u, u_x, u_y, u_{xy}\}$. (b) Master–slave Bloch coupling at periodic boundaries. The phase $e^{i\mathbf{k} \cdot \mathbf{a}\alpha}$ applies to displacement and derivative degrees of freedom alike.

Reduced Hermitian eigenproblem.. We obtain the dispersion curves $\bar{\omega}_n(\mathbf{k})$ from the reduced generalized eigenvalue problem:

$$[\bar{\mathbf{K}}(\mathbf{k}) - \omega^2 \bar{\mathbf{M}}(\mathbf{k})] \bar{\mathbf{d}} = \mathbf{0}. \quad (40)$$

Each entry of the diagonal phase subblock $\mathbf{T}_{SM}(\mathbf{k})$ has unit modulus, so this subblock is unitary. The full map $\mathbf{T}(\mathbf{k})$ is rectangular and full-column-rank, but is not unitary. For positive elastic and inertial parameters, the full-cell matrices satisfy $\mathbf{K} \succeq 0$ and $\mathbf{M} \succ 0$; the reduced operators are therefore Hermitian:

$$\bar{\mathbf{K}}(\mathbf{k})^H = \bar{\mathbf{K}}(\mathbf{k}) \succeq 0, \quad \bar{\mathbf{M}}(\mathbf{k})^H = \bar{\mathbf{M}}(\mathbf{k}) \succ 0, \quad (41)$$

The generalized eigenvalues ω_n^2 are therefore real and nonnegative. Shifting $\mathbf{k}$ by a reciprocal-lattice vector $\mathbf{G}$ leaves every Bloch phase unchanged, so the reduced stiffness and mass matrices satisfy $\bar{\mathbf{K}}(\mathbf{k} + \mathbf{G}) = \bar{\mathbf{K}}(\mathbf{k})$ and $\bar{\mathbf{M}}(\mathbf{k} + \mathbf{G}) = \bar{\mathbf{M}}(\mathbf{k})$. The spectrum is periodic across reciprocal space.

Branch tracking.. Modes with different spatial symmetries can cross along high-symmetry paths. We track their identity between adjacent wave vectors using the Modal Assurance Criterion (MAC) at $\mathbf{k}_1$ and $\mathbf{k}_2$. Let $\mathbf{d}_m(\mathbf{k}) = \mathbf{T}(\mathbf{k})\bar{\mathbf{d}}_m(\mathbf{k})$ be the corresponding full-cell eigenvector, and let $\mathbf{M}$ be the common, assembled full-cell mass matrix, including gradient micro-inertia. The cross-wavevector MAC is

$$\text{MAC}_{mn}(\mathbf{k}_1, \mathbf{k}_2) = \frac{|\mathbf{d}_m(\mathbf{k}_1)^H \mathbf{M} \mathbf{d}_n(\mathbf{k}_2)|^2}{(\mathbf{d}_m(\mathbf{k}_1)^H \mathbf{M} \mathbf{d}_m(\mathbf{k}_1)) (\mathbf{d}_n(\mathbf{k}_2)^H \mathbf{M} \mathbf{d}_n(\mathbf{k}_2))}. \quad (42)$$

The lifted vectors use a common full-cell mass metric; the reduced mass matrix depends on $\mathbf{k}$ and is not used to compare modes at different wave vectors. If the off-diagonal MAC indicates a branch exchange, we halve the continuation step. We compute band gaps from the frequency-ordered eigenvalues $\bar{\omega}_1(\mathbf{k}) \leq \bar{\omega}_2(\mathbf{k}) \leq \dots \leq \bar{\omega}_N(\mathbf{k})$, independently of the mode labels assigned during tracking.

5. Verification and numerical resolution

We compare the implementation with analytical limits, independent transfer-matrix calculations, algebraic identities, and mesh-refinement results. For the published heterogeneous bilayers, we report only qualitative graphical agreement where source curves are available. Cases with ambiguous parameters or no numerical curves are not treated as external validation.

5.1. Analytical limits and bilayer comparison

In the isotropic homogeneous limit, the long-wavelength phase velocities recover the classical transverse and longitudinal speeds:

$$\lim_{k \rightarrow 0} \frac{\omega_T(k)}{k} = \sqrt{\frac{\mu}{\rho}} = 1.0, \quad \lim_{k \rightarrow 0} \frac{\omega_L(k)}{k} = \sqrt{\frac{\lambda + 2\mu}{\rho}} = \sqrt{3} \approx 1.732 \quad (43)$$

At $\bar{k} = 0.001$, the computed acoustic speeds differ from these limits by 1.25×10^{-8} (Table 1). The pure-bending micro-beam reduction matches the published closed-form

limits to within 10^{-15} [24]; details appear in Supplementary Material S2. The micro-inertia asymptote is checked separately in Appendix A and Section 7.

As a classical reference, we analyze a periodic AlN/BaTiO₃ bilayer with $a_A = a_B = 0.01$ m and $b = a_A + a_B = 0.02$ m using the Rytov relation [30, 28]:

$$\cos(Kb) = \cos(k_A a_A) \cos(k_B a_B) - \frac{1}{2} \left(\frac{Z_A}{Z_B} + \frac{Z_B}{Z_A} \right) \sin(k_A a_A) \sin(k_B a_B), \quad (44)$$

where $k_m = \omega \sqrt{\rho_m / c_{33,m}}$, $v_m = \sqrt{c_{33,m} / \rho_m}$, and $Z_m = \rho_m v_m$. The single-layer transfer-matrix identity and the identical-material bilayer reduction match their analytical limits to 6.47×10^{-16} and 7.22×10^{-16} , respectively. The heterogeneous transfer-matrix trace reproduces the Rytov relation to 2.22×10^{-16} . The first three calculated stop-band intervals are $[0.4806, 0.5209]$, $[0.9812, 1.0215]$, and $[1.4993, 1.5028]$ in normalized frequency. We compare the result with the published B1 plot visually because tabulated source data are unavailable; no percentage error is inferred from the curves.

We reproduce two gradient-elastic bilayer cases, but do not treat them as external validations. One published length-scale definition is dimensionally ambiguous. For the other, the plotted parameter set and normalization are unspecified, and the numerical curves are unavailable. Supplementary Material S2 records these limitations and the status of each comparison. We also compare the dipolar layer matrix with the source’s printed formulation; this checks the formulation but does not validate its band curves.

5.2. Numerical consistency and mesh convergence

The consistency checks assess operator Hermiticity, Brillouin-zone periodicity, isotropic rotational invariance, acoustic slopes, rotation–swap symmetry, matrix definiteness, the micro-inertia limit, and the energy-flux/group-velocity identity. Table 1 reports the values obtained for these comparisons. After the harmonic micro-inertia sign correction in Appendix Appendix B, the energy-flux residual cannot be independently recalculated because the raw harmonic fields are unavailable. We therefore retain this identity as an unresolved diagnostic rather than treating the reported residual as independent confirmation. Supplementary Material S3 gives the check definitions and expanded diagnostics.

We assess mesh convergence for homogeneous Case H on uniform $N \times N$ BFS grids, $N \in \{4, 8, 16, 32\}$, at fixed $\mathbf{k} = (0.31\pi/L, 0.22\pi/L)$. Figure 4 shows the acoustic-frequency error relative to the closed-form value and the change between successive meshes. The fitted empirical rate is $p = 4.17$ (95% CI $[3.15, 5.20]$) over the 4^2 , 8^2 , and 16^2 levels. The reported 32^2 result is resolution-limited and is excluded from the fit. This rate is empirical; it is not a theoretical-order claim. The operational mesh-change floor is $\varepsilon_\Delta = 4.63 \times 10^{-11}$ and is distinct from

eigensolver reproducibility. The smallest reported directional gap, $|\Delta_{GX}| = 4.3 \times 10^{-3}$, exceeds the floor by approximately 9×10^7 .

Selected Case-H and Case-C inputs appear with the Results. Supplementary Material S1–S4 contains the full parameter summary, external-comparison details, and extended convergence diagnostics.

6. Results: dispersion, band gaps, and wave steering

We proceed from the homogeneous isotropic reference to tensor-driven branch shifts, aspect-ratio and orientation effects, directional stop bands and their sampled design map, and group-velocity steering. We then treat the heterogeneous Case C separately because it introduces periodic material contrast. Table 2 lists selected inputs; Supplementary Material S1 gives the full parameter set.

6.1. Homogeneous isotropic baseline

At $AR = 1$, the principal characteristic lengths coincide and the rotated length tensor is isotropic, so its constitutive response is independent of θ . This case provides the reference for the anisotropic calculations. Its reported Γ – X directional width is $\Delta_{GX} = 0.041$; this path-specific value is not a complete-gap measure. Figure 5 includes the isotropic branches for comparison with the anisotropic cases.

6.2. Anisotropy reshapes the Case-H branches

Figure 5 compares the isotropic reference with two orientations of the anisotropic length tensor. Both acoustic branches start at zero frequency at Γ . Changing aspect ratio and orientation shifts the branches and lifts symmetry-related degeneracies at the zone edge. At $AR = 10$ and $\theta = 45^\circ$, a narrow stop band appears along Γ – X , with $\Delta_{GX} = 0.0043$. Modes propagating in other directions overlap this interval, so it is not a complete gap; the full-zone measure remains $\Delta_{\text{complete}} \leq -0.3235$.

In this model, the rotated length tensor changes the gradient-stiffness contribution according to propagation direction. The resulting branch shifts are therefore not uniform across the Brillouin zone: adjacent bands can separate along Γ – X while modes at other wave vectors remain in the same frequency interval.

6.3. Aspect-ratio and orientation effects

Figure 6 shows the zone-edge frequencies as the tensor rotates. At $AR = 5$, as θ changes from 0° to 90° , the transverse and longitudinal frequencies at X fall from 2.915 to 2.672 and from 5.050 to 4.628, respectively, as the projected length scale shifts from the major to the minor axis. The M -point frequencies are symmetric about 45° , as expected from the square-cell symmetry.

Figure 7 isolates elongation at fixed $\theta = 45^\circ$, using volume-equivalent length scaling. Increasing AR lengthens

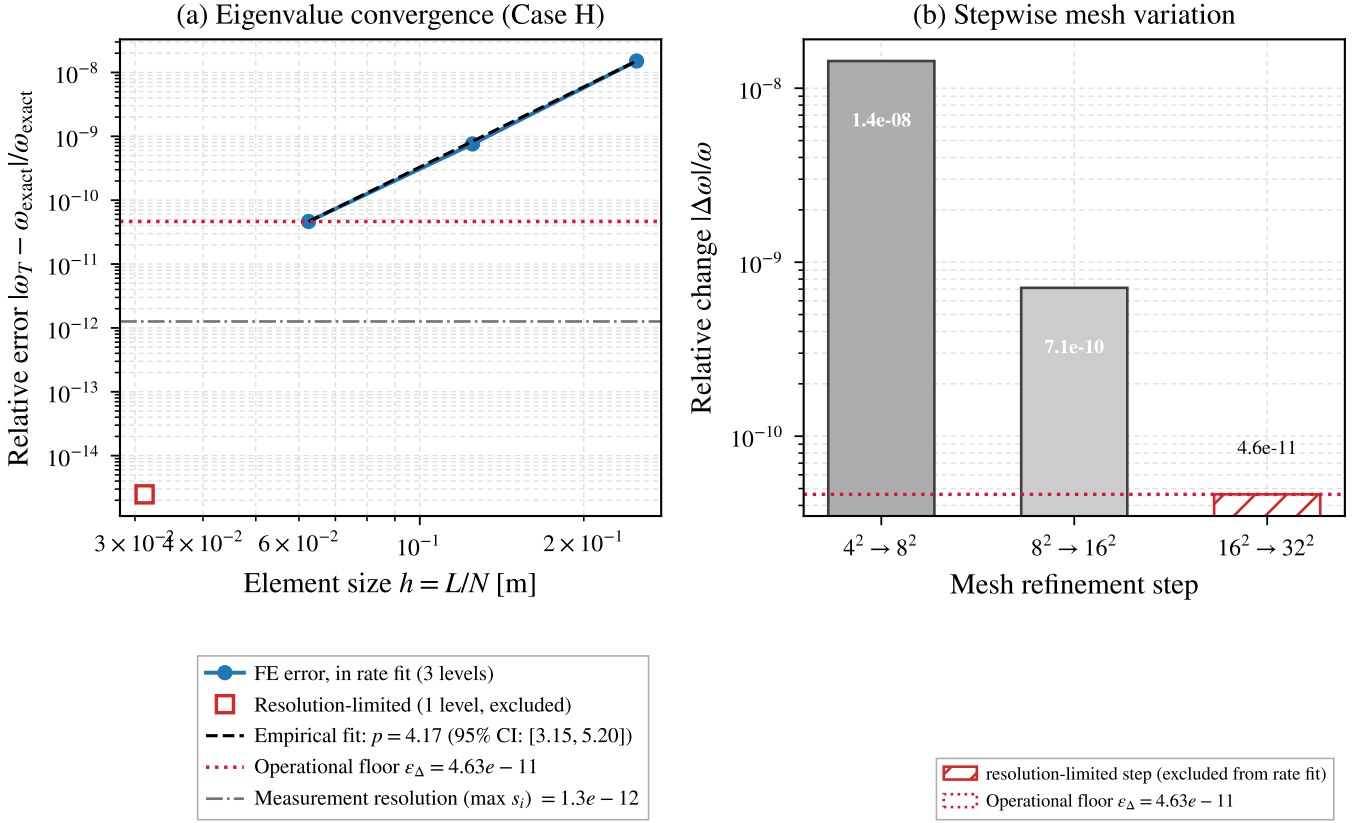


Figure 4: Case-H mesh study for the acoustic branch. (a) Relative frequency error versus element size; the 32^2 result is shown as resolution-limited and is not included in the empirical fit. (b) Successive-mesh change and the operational mesh-change floor ε_Δ .

Table 1: Numerical consistency checks for the Bloch finite-element implementation.

Check	Quantity assessed	Maximum discrepancy	Criterion	Result
Hermiticity	Reduced stiffness and mass operators	2.01×10^{-16} ; 5.63×10^{-17}	$< 10^{-12}$	Within criterion
Brillouin-zone periodicity	Spectrum under reciprocal-lattice shifts	9.66×10^{-16}	$< 10^{-10}$	Within criterion
Isotropic rotation	Frequencies at $\text{AR} = 1$	9.63×10^{-16}	$< 10^{-10}$	Within criterion
Acoustic limit	Long-wave transverse and longitudinal slopes	1.25×10^{-8}	$< 10^{-6}$	Within criterion
Rotation-swap symmetry	Spectrum under $\theta \mapsto \theta + 90^\circ$ and axis swap	6.44×10^{-16}	$< 10^{-10}$	Within criterion
Definiteness	Two rigid modes at Γ ; positive interior spectrum	As expected	Admissible	As expected
Micro-inertia limit	Phase speed at $\bar{k} = 200$ versus asymptotic value	2.85×10^{-4}	$< 10^{-3}$	Within criterion
Energy-flux identity	Group velocity versus cycle-averaged flux	6.91×10^{-10}	$< 10^{-6}$	Unresolved

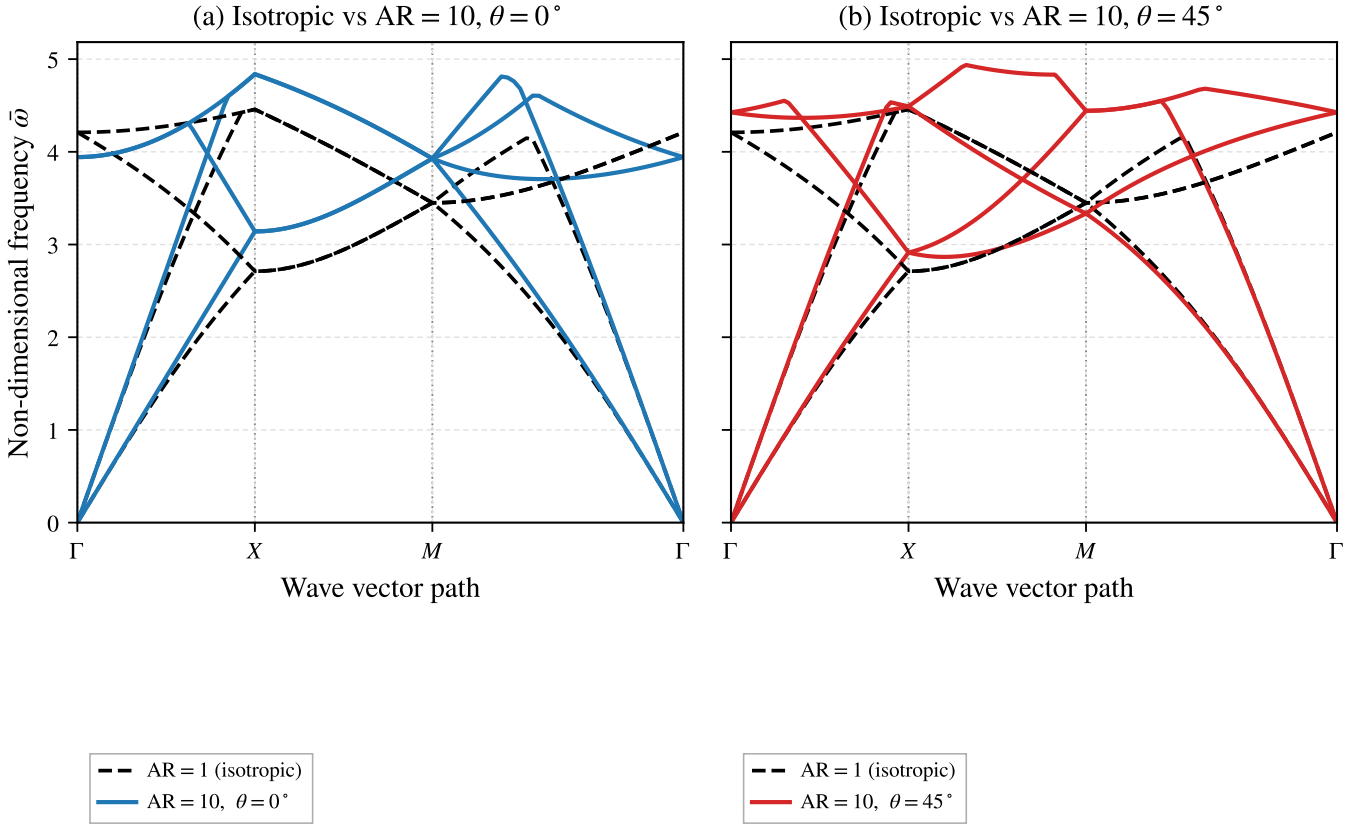
the major characteristic length while shortening the minor one, so the directional gradient-stiffness contribution and the two bounding band edges need not shift together. Because the gap width is their frequency difference, it can vary non-monotonically even when an individual branch moves consistently.

6.4. Directional stop bands

A positive Δ_{GX} identifies a stop band on the Γ - X leg only. It does not imply that the same interval is empty of modes over the full Brillouin zone. Thus, although the $\text{AR} = 10$, $\theta = 45^\circ$ case has $\Delta_{GX} = 0.0043$, its complete-gap measure is negative. Every listed homogeneous con-

Table 2: Selected material and numerical inputs for Cases H and C and the steering analysis.

Case	Quantity	Value
H	Cell and matrix	$L = 1.0 \text{ m}; \lambda = \mu = 1.0 \text{ Pa}; \rho = 1.0 \text{ kg/m}^3$
H	Lengths and mesh	$l_{\text{iso}} = 0.20 \text{ m}; \ell_i = 0.20 \text{ m}; 16 \times 16 \text{ BFS elements per cell}$
C	Epoxy matrix	$\rho_m = 1142 \text{ kg/m}^3; \mu_m = 1.48 \text{ GPa}; \lambda_m = 4.57 \text{ GPa}$
C	YBCO inclusion	$\rho_i = 6333 \text{ kg/m}^3; \mu_i = 37.0 \text{ GPa}; \lambda_i = 95.0 \text{ GPa}$
C	Geometry and length scales	$L = 1.0 \text{ m}; r_0/L = 0.30; \ell_m/L = 0.10; \ell_i/L = 0.20$
C	Frequency scale and reported resolution	$c_{t,m} = 1138.4 \text{ m/s}; \omega_{0,C} = \pi c_{t,m}/L = 3576.4 \text{ rad/s}; 4 \times 4 \text{ mesh and } 21 \times 21 \text{ zone grid}$
Steering	Acoustic-branch sweep	$\bar{k} = 0.5; \text{AR} \in \{5, 10\}; \theta = 0^\circ, 15^\circ, \dots, 90^\circ$


 Figure 5: Case-H Bloch bands along Γ -X-M- Γ . (a) Isotropic reference (AR = 1) and aligned anisotropy (AR = 10, $\theta = 0^\circ$). (b) Isotropic reference and diagonal anisotropy (AR = 10, $\theta = 45^\circ$), which opens a Γ -X stop band but no complete gap.

figuration likewise has a negative complete-gap value. Table 3 reports the representative directional and complete-gap measures; Supplementary Material S3 gives normalized widths and mesh-to-mesh variations.

6.5. Design-map interpretation

Across the 42 evaluated (θ, AR) configurations in Figure 8, the largest sampled $\Delta_{GX} = 0.0677$ occurs at $(0^\circ, 7)$. The response is non-monotone in aspect ratio. The plotted

surface interpolates the evaluated points; it is not a continuous optimization. Figure 9 classifies the same samples by the sign of Δ_{GX} , so its regime boundary is a visual summary of the sampled response. The orientation sensitivity S_θ is largest at AR = 10 (2.34 rad^{-1}).

6.6. Group-velocity steering

On the constant-wavenumber ring $|\mathbf{k}| = \bar{k} = 0.5$, anisotropy makes the group velocity non-collinear with

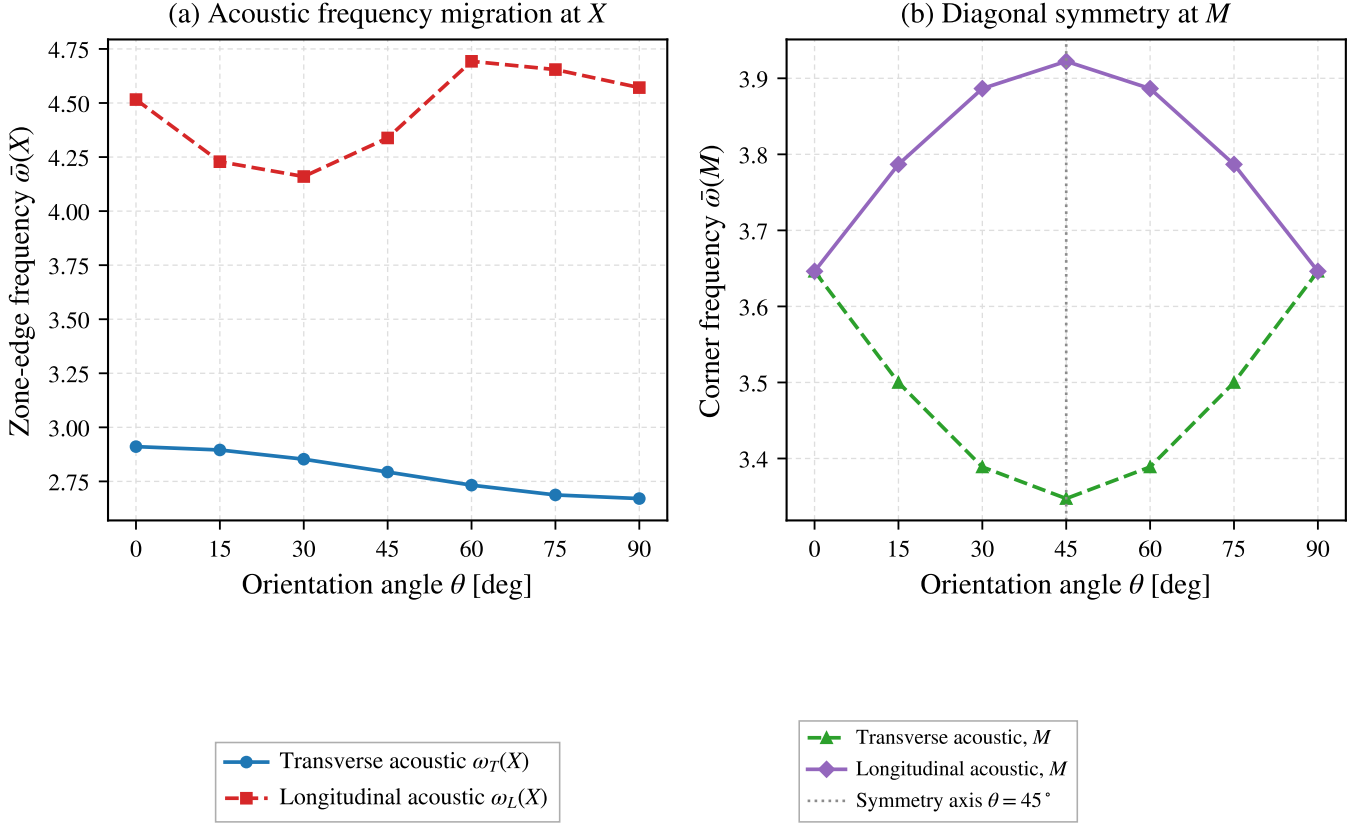


Figure 6: Orientation dependence of the first four Case-H eigenfrequencies at (a) X and (b) M for $AR = 5$. The M -point response is symmetric about $\theta = 45^\circ$.

Table 3: Directional and complete-gap measures for representative Case-H configurations; values are reported at the precision retained between the 8×8 and 16×16 meshes.

AR	θ	Δ_{GX}	Δ_{path}	Δ_{complete}
1	0°	+0.041	-0.32	-0.32
1	45°	+0.041	-0.32	-0.32
1	90°	+0.041	-0.32	-0.32
3	0°	+0.045	-0.10	-0.46
3	45°	+0.044	-0.45	-0.48
3	90°	+0.011	-0.33	-0.46
5	0°	+0.001	-0.01	-0.58
5	45°	+0.016	-0.48	-0.58
5	90°	+0.022	-0.37	-0.58
10	0°	-0.002	-0.24	-0.85
10	45°	+0.004	-0.29	-0.56
10	90°	+0.031	-0.65	-0.85

S_θ [rad $^{-1}$]: AR = 1: 0.00; 2: 1.42; 3: 1.76; 5: 1.86; 7: 1.40; 10: 2.34.

the wave vector. Since $\mathbf{v}_g = \nabla_{\mathbf{k}}\omega$ is normal to the local constant-frequency contours, their slope determines the angular departure of $\mathbf{v}_g$ from $\mathbf{k}$. Figure 10 plots the angular deviation $\delta(\phi) = \angle \mathbf{v}_g - \angle \mathbf{k}$ and group-velocity magnitude at $\theta = 45^\circ$. The peak deviation rises from 0° at $AR = 1$ to 1.4146° at $AR = 5$ and 2.7846° at $AR = 10$.

The magnitude $|\mathbf{v}_g|$ also varies around the contour, so the anisotropic dispersion changes the energy-transport speed as well as its direction.

Figure 11 and Table 4 report the orientation sweep $\theta \in \{0^\circ, 15^\circ, \dots, 90^\circ\}$ for $AR = 5$ and 10 on the same ring and acoustic branch. Across the sampled orientations,

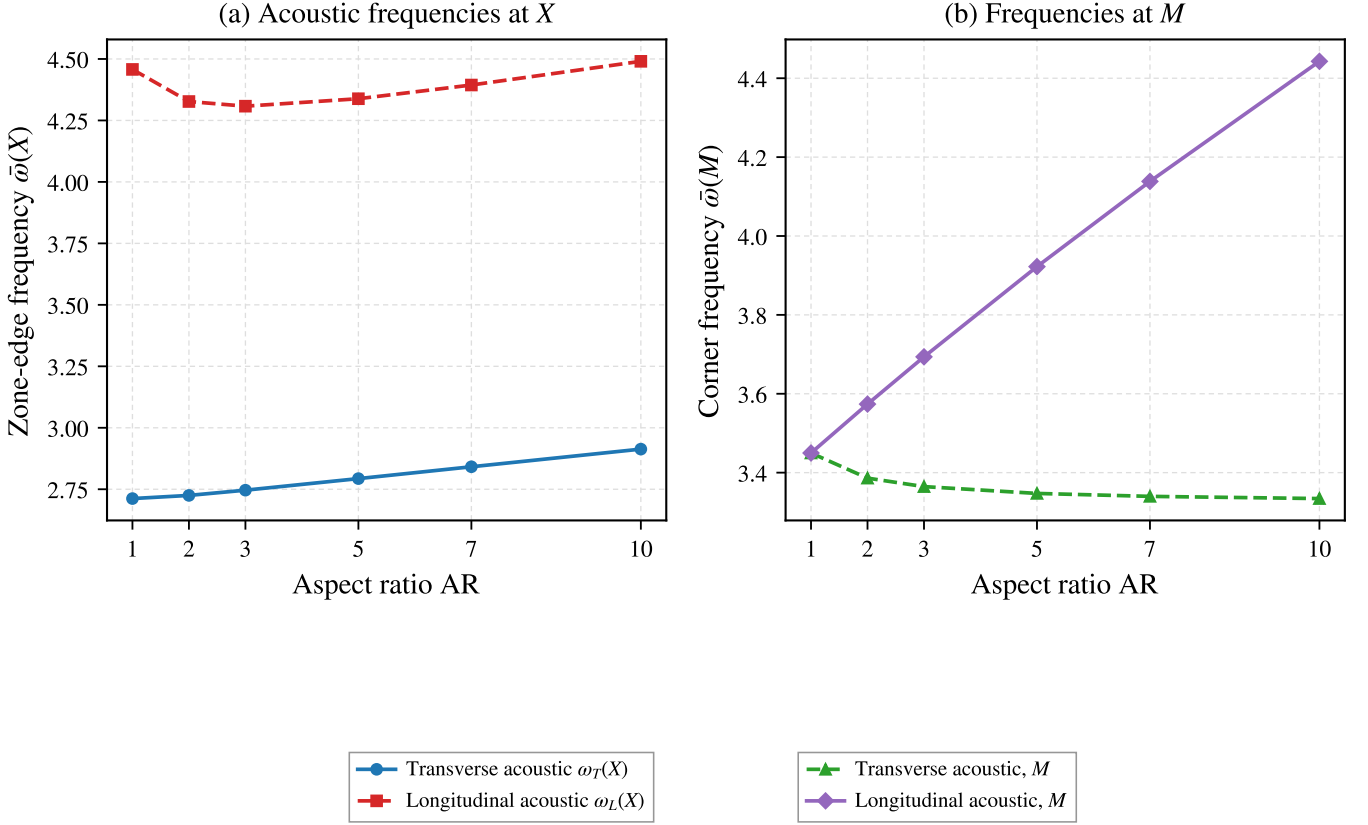


Figure 7: Case-H branch frequencies at X and M as the aspect ratio varies from 1 to 10 at fixed $\theta = 45^\circ$, using volume-equivalent length scaling.

the peak deviation remains 1.4146° and 2.7846° , respectively, to within 10^{-5} deg. The principal direction follows $\phi^*(\theta) \simeq (45^\circ - \theta) \bmod 90^\circ$; at $AR = 10$, its excursions are at most 5° . These samples show that rotating the tensor repositions the steering lobes while leaving the peak angular deviation nearly unchanged; for this branch and ring, orientation chiefly redirects the energy flow rather than increasing its sampled maximum.

We define the sampled steering measure as follows:

$$M_s(AR) = \phi^*(90^\circ, AR) - \phi^*(0^\circ, AR), \quad (45)$$

For the two aspect ratios, the values are:

$$M_s(AR = 5) = 0^\circ, \quad M_s(AR = 10) = 10^\circ. \quad (46)$$

The 5° ring-sampling step sets the resolution of this difference. The deviation field is headless-periodic and mirror co-symmetric, with residuals no greater than 1.3×10^{-6} deg. The peak amplitude is invariant under $\theta \mapsto 90^\circ - \theta$ to within 10^{-7} deg. These symmetries are consistent with orientation shifting the lobes while leaving the sampled peak deviation unchanged.

6.7. Heterogeneous Case C: complete gap with periodic contrast

Case C combines gradient elasticity with a periodic epoxy–YBCO impedance contrast ($\chi_\mu = 25.0$, $\chi_\rho = 5.546$;

Table 2), unlike homogeneous Case H. We denote the cell period by L in the text and a in Figure 12; $r_0/L = r_0/a = 0.30$ gives a filling fraction of 28.27%. Figure 12 shows a complete gap between Bands 3 and 4 over $\bar{\omega} \in [4.5698, 7.1430]$. On the 4×4 mesh and 21×21 Brillouin-zone grid, $\Delta_{\text{complete}} = 2.5732$, or 43.94%. We integrate the circular interface by sub-cell Gauss sampling on the Cartesian mesh. Brillouin-zone refinement changes the quoted width by less than 4×10^{-4} ; finite-element refinement has a much larger effect.

Under mesh refinement, the complete gap falls from 2.5732 at 4×4 to 2.2504, 2.0722, 1.9736, and 1.9179 at 8×8 , 16×16 , 32×32 , and 64×64 , respectively. It stays open at every computed resolution but has not converged. The value 2.5732 is therefore a coarse-mesh result, not a continuum estimate. At the 4×4 mesh, increasing r_0/L from 0.20 to 0.30 widens the gap from 0.8288 (16.61%) to 2.5732 (43.94%); it then narrows to 2.4426 (37.00%) at $r_0/L = 0.40$. Supplementary Material S4 gives the mesh, quadrature, and zone-sampling details.

7. Gradient micro-inertia and the high-wavenumber response

Gradient micro-inertia adds kinetic energy associated with spatially varying acceleration. This section exam-

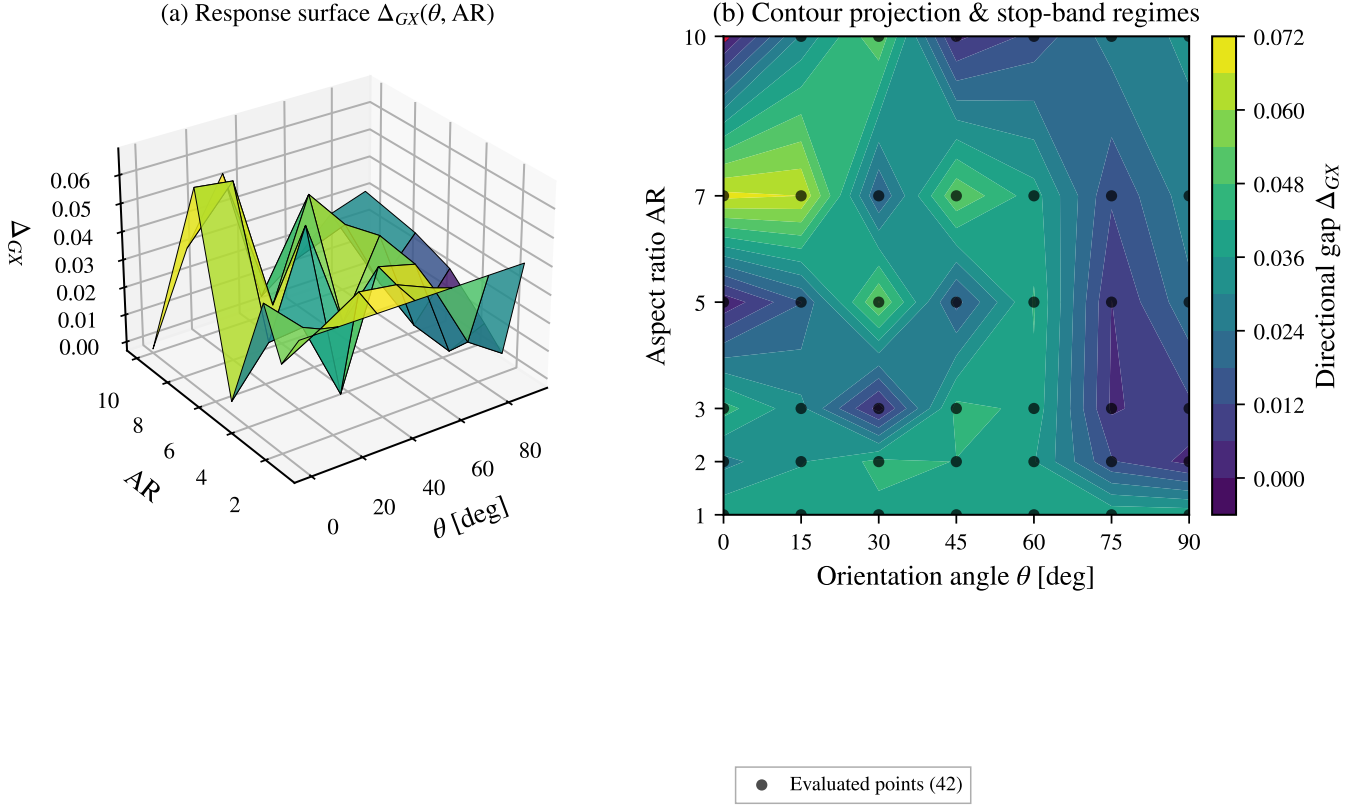


Figure 8: Sampled Case-H design response $\Delta_{GX}(\theta, AR)$ for 42 configurations. The largest sampled value is 0.0677 at $\theta = 0^\circ$, $AR = 7$; at $\theta = 45^\circ$, $AR = 10$, $\Delta_{GX} = 0.0043$.

Table 4: Peak group-velocity deviation and principal steering direction across the sampled orientations at $\bar{k} = 0.5$.

θ	AR = 5		AR = 10	
	$\delta_{\max}$ [deg]	ϕ^* [deg]	$\delta_{\max}$ [deg]	ϕ^* [deg]
0°	1.4146	45	2.7846	40
15°	1.4146	30	2.7846	35
30°	1.4146	15	2.7846	20
45°	1.4146	0	2.7846	5
60°	1.4146	75	2.7846	70
75°	1.4146	60	2.7846	55
90°	1.4146	45	2.7846	50

$M_s = \phi^*(90^\circ) - \phi^*(0^\circ)$: $M_s(AR = 5) = 0^\circ$ and $M_s(AR = 10) = 10^\circ$. The 5° ring-sampling step limits the resolution of this difference; parabolic peak refinement remains within one sampling step.

Headless periodicity and mirror co-symmetry residuals are each $\leq 1.3 \times 10^{-6}$ deg: $\delta(\phi; \theta) = \delta(\phi + 180^\circ; \theta)$ and $\delta(\phi; \theta) = \delta(90^\circ - \phi; 90^\circ - \theta)$.

ines its contribution to the strain-energy partition and the high-wavenumber response of the transverse shear branch studied here.

7.1. Energy partition and the transverse-branch limit

For Mindlin Form-II elasticity, the strain energy splits into classical and gradient parts, $W = W_c + W_g$, with $W_c = \frac{1}{2} \int_{\Omega} \sigma_{ij} \varepsilon_{ij} d\Omega$ and $W_g = \frac{1}{2} \int_{\Omega} \tau_{ijk} \eta_{ijk} d\Omega$. Along $\Gamma - X$, the gradient-energy fraction rises from 0.20% at $\bar{k} = 0.1$

to 16.49% at $\bar{k} = 1.0$ (Figure 13). Double-stress work therefore contributes more as the wavelength approaches the cell scale.

The transverse shear branch analyzed in Appendix Appendix A has the asymptotic limit

$$\bar{v}_{T,\infty} = \frac{\bar{l}_{\text{eff}}}{\sqrt{10} \bar{\ell}_i} = \frac{0.20}{\sqrt{10} \times 0.20} = \frac{1}{\sqrt{10}} \approx 0.3162. \quad (47)$$

At $\bar{k} = 200$, the computed phase speed with micro-

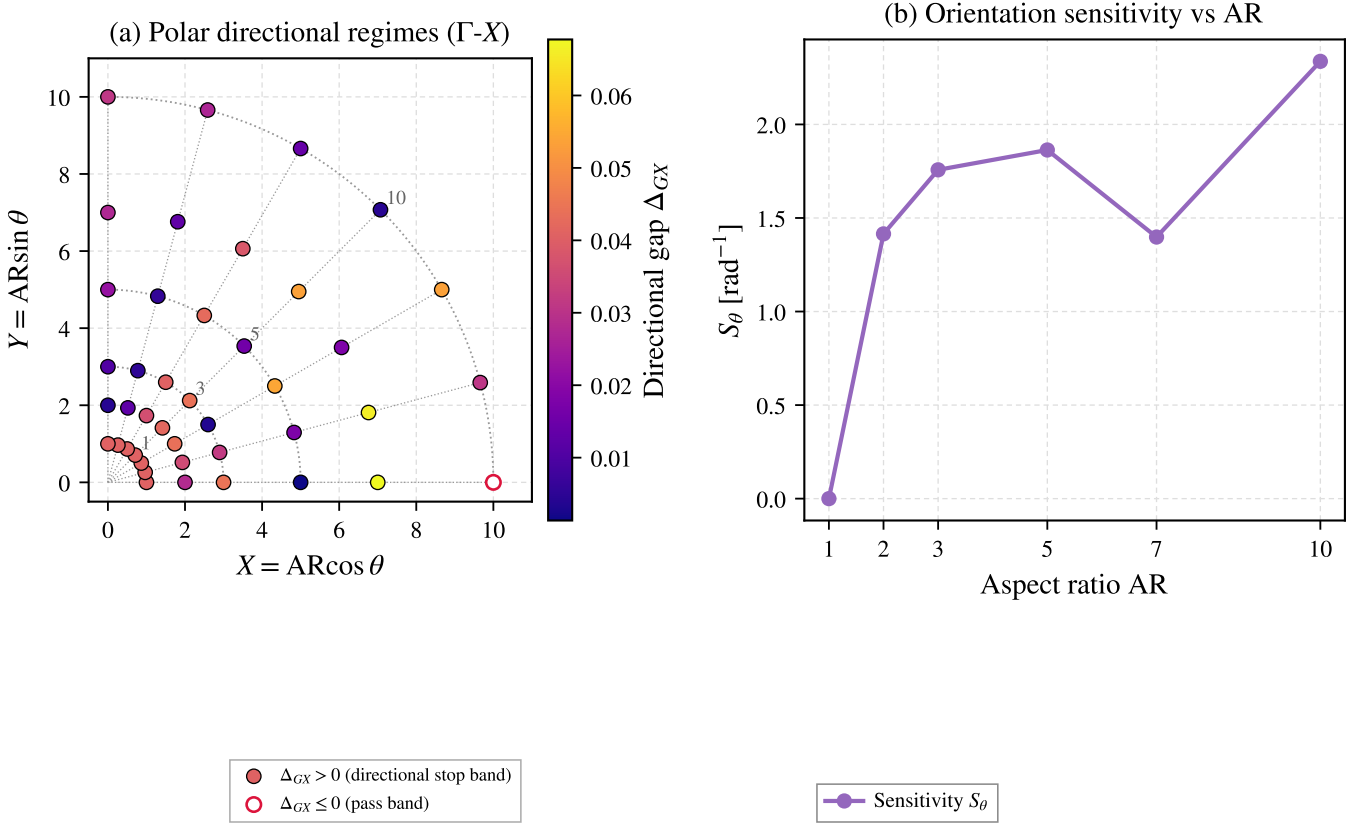


Figure 9: Polar-coordinate view of the sampled Case-H designs, classified by $\Delta_{GX} \leq 0$ (pass band) and $\Delta_{GX} > 0$ (directional stop band). The displayed boundary is a visual summary of sampled points.

inertia is 0.3163, within 0.03% of this asymptote. Without micro-inertia, the analytical branch relation grows without bound, and the computed phase speed reaches 39.75 at $\bar{k} = 200$. In Eq. (A.4), the gradient-stiffness and micro-inertia corrections to v_p^2 both grow as k^2 ; their ratio sets the finite high-wavenumber limit, while setting $\ell_i = 0$ removes the inertial denominator term and restores the unbounded trend. This result applies only to the transverse branch and constitutive model studied; it does not establish a speed bound for other branches or models.

8. Discussion and limitations

In homogeneous Case H, the rotated characteristic-length tensor changes the gradient-stiffness contribution according to the wave-vector direction relative to its principal axes. The resulting branch shifts are unequal: rotating the tensor can separate neighboring bands along Γ -X while modes at other wave vectors remain in the same frequency interval. Under the volume-equivalent scaling, changing the aspect ratio lengthens one principal characteristic length while shortening the other, so the two bounding band edges need not move together. Their frequency difference can therefore vary non-monotonically. A positive Δ_{GX} is a stop band only along Γ -X; modes else-

where in the Brillouin zone remain consistent with the negative complete-gap measure $\Delta_{\text{complete}} \leq -0.3235$. Thus, the homogeneous design supports direction-selective filtering, not full-zone attenuation. The 42-point map can compare evaluated configurations but does not identify a continuous optimum.

The group-velocity response follows from the same direction-dependent dispersion. Because $\mathbf{v}_g = \nabla_{\mathbf{k}} \omega$ is normal to a local constant-frequency contour, changes in contour slope alter the angle between the energy-flow direction and the wave vector. Rotating the length tensor reorients these local contour normals and shifts the steering lobes. On the sampled acoustic branch at $\bar{k} = 0.5$, the peak deviation is nearly unchanged while its principal direction moves. Within this calculation, orientation therefore redirects the energy flow without materially increasing its peak angular deviation; this interpretation is limited to the branch, ring, and sampled orientations.

Case C adds periodic contrast in both shear modulus and density. The resulting Bloch bands separate Bands 3 and 4 over the sampled Brillouin zone, unlike the path-specific branch separation in homogeneous Case H. This difference is consistent with the positive complete-gap measure in Case C and the absence of a complete gap in Case H. The Case-C gap decreases from 2.5732 on the 4×4

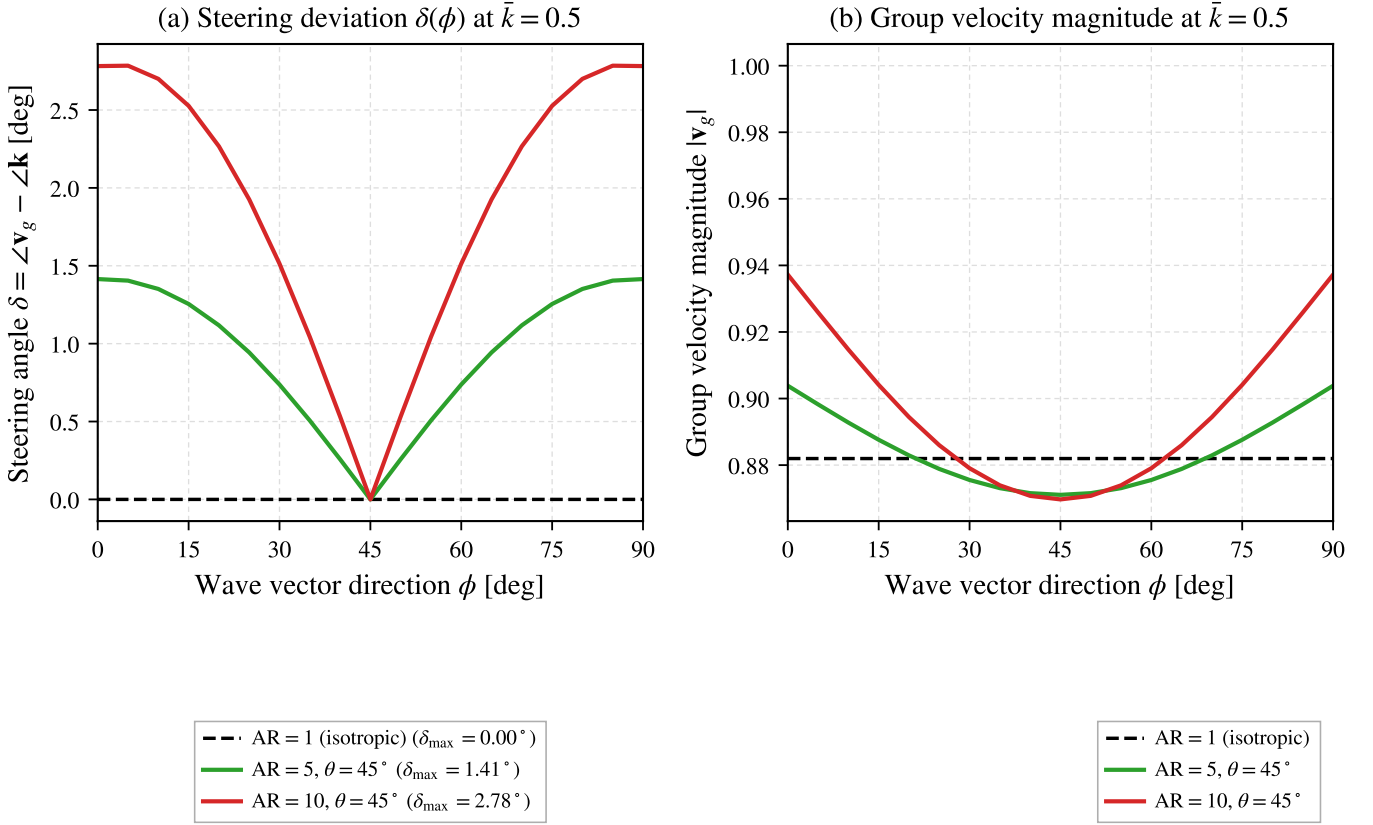


Figure 10: Acoustic-branch group-velocity steering on $|\mathbf{k}| = 0.5$ at $\theta = 45^\circ$. (a) Angular deviation for AR = 1, 5, 10, with peak values 0° , 1.4146° , and 2.7846° . (b) Directional variation of group-velocity magnitude.

mesh to 1.9179 on the 64×64 mesh and remains unconverged. Because the cases use different material and discretization settings, the comparison is interpretive rather than a controlled parameter sweep; the coarse-mesh Case-C width is not a continuum estimate.

The homogeneous mesh study gives an empirical rate $p = 4.17$ over the 4^2 , 8^2 , and 16^2 levels. The 32^2 result is resolution-limited and excluded from the fit, so this rate is not a theoretical-order claim. Seven of the eight tabulated numerical consistency checks meet their stated criteria. After the harmonic micro-inertia sign correction in Appendix Appendix B, the energy-flux/group-velocity identity remains unresolved because the raw harmonic fields needed for an independent recalculation are unavailable.

The classical AlN/BaTiO₃ calculation reproduces the Rytov transfer-matrix relation to 2.22×10^{-16} ; comparison with the published B1 curves is visual only because tabulated source values are unavailable. The two gradient-elastic bilayer cases are reproductions, not quantitative external validations: source parameter conventions are unresolved and the numerical curves needed for comparison are unavailable. No quantitative validation claim is made for those heterogeneous benchmarks.

For the transverse branch in Appendix Appendix A, gradient stiffness raises the phase speed at high wavenumber

when micro-inertia is absent. With micro-inertia, the k^2 contribution in the numerator of v_p^2 is balanced asymptotically by the k^2 inertial term in its denominator, leaving a finite limit set by the ratio of the gradient-stiffness and micro-inertia lengths. This is a mechanism-specific result for the branch and constitutive model studied, not a universal bound. More broadly, the model is linear, local, isothermal, and two-dimensional under plane-strain assumptions; it excludes nonlinear gradient effects, dissipation, piezoelectric coupling, and three-dimensional polarization modes.

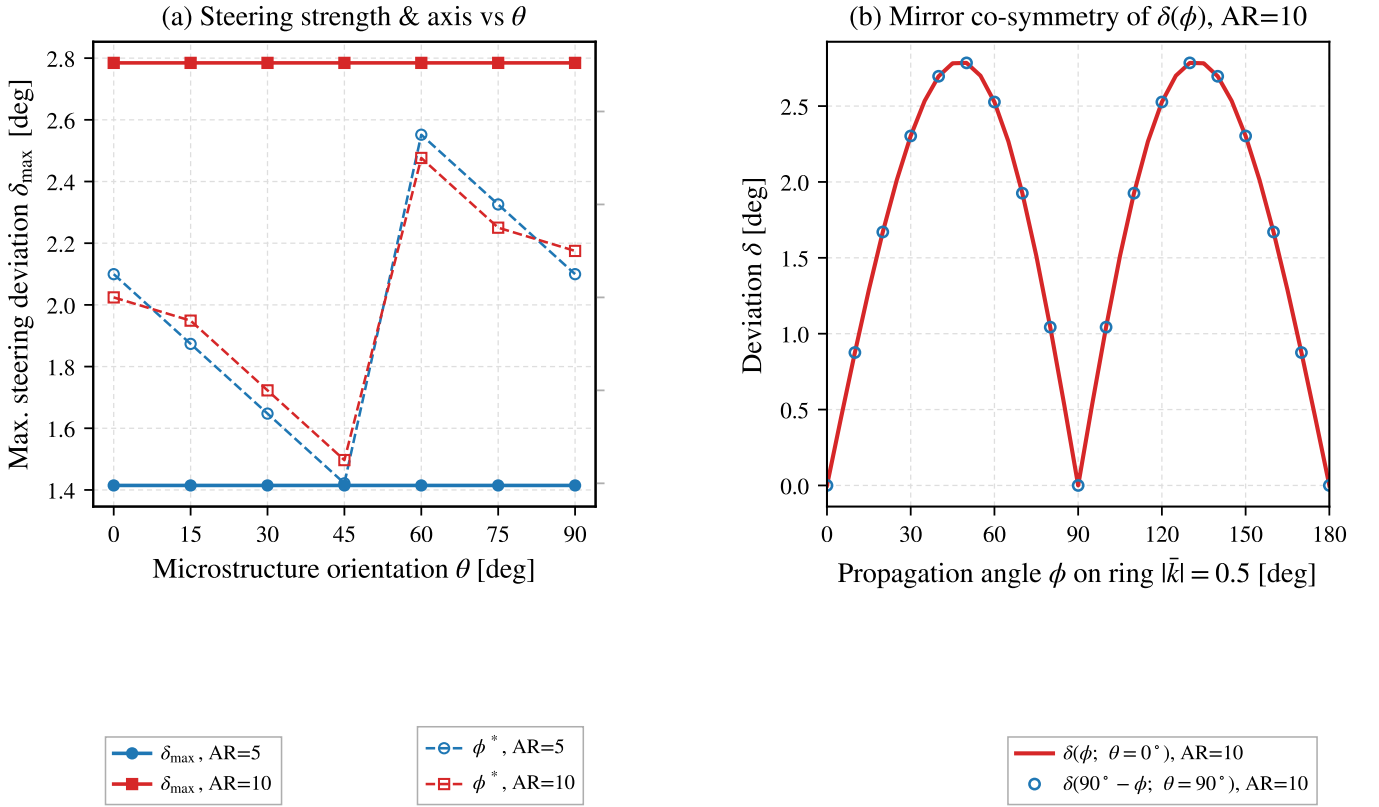


Figure 11: Orientation dependence of steering on $|\mathbf{k}| = 0.5$. (a) Peak deviation $\delta_{\max}$ and principal direction ϕ^* for AR = 5, 10; the peak angular deviation is invariant over the sampled orientations, while the lobe direction follows the rotated tensor axes on the 5° ring grid. (b) The $\theta = 0^\circ$ and 90° deviation fields at AR = 10 exhibit mirror co-symmetry.

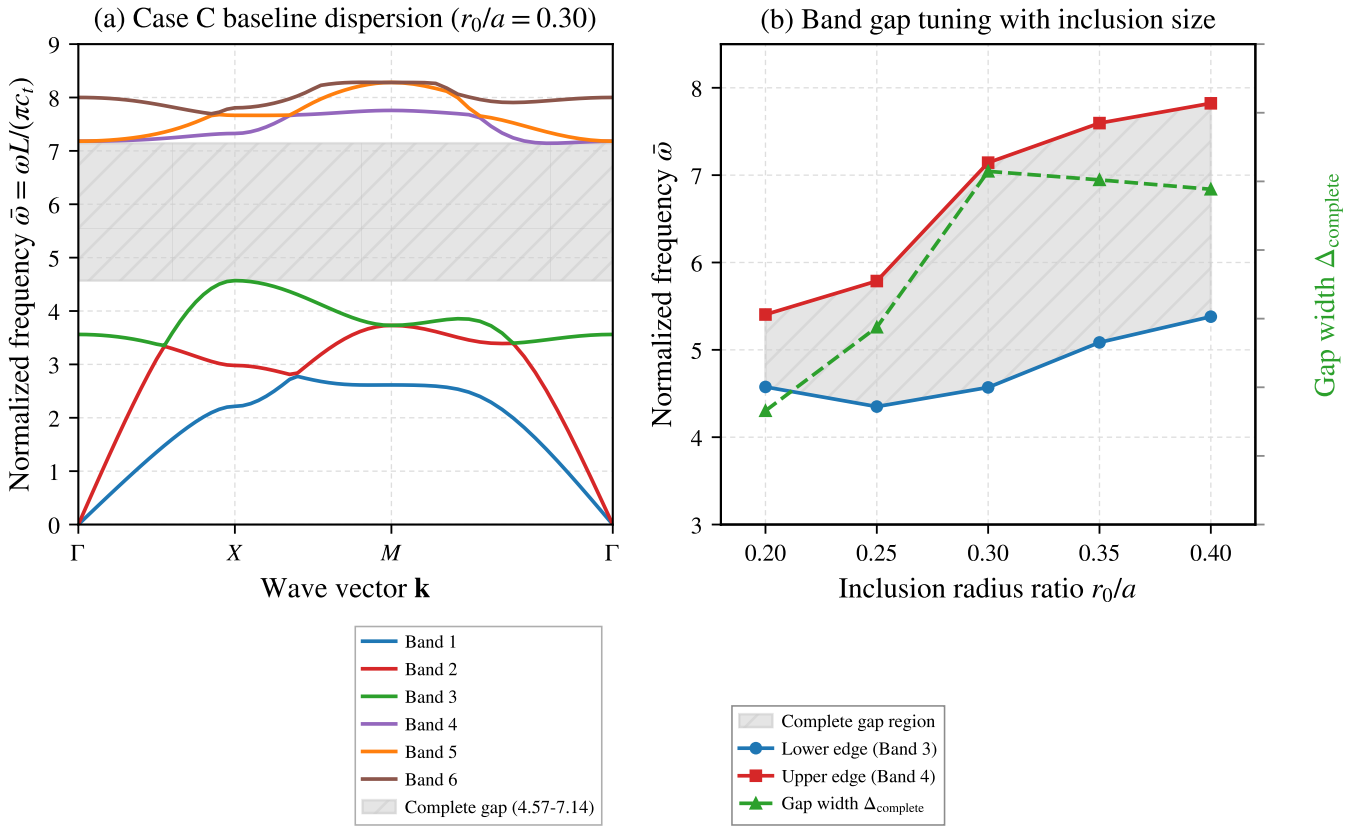


Figure 12: Case-C complete-gap formation and radius dependence. (a) Bands along Γ - X - M - Γ for $r_0/L = 0.30$, with the 4×4 -mesh gap between Bands 3 and 4. (b) Complete-gap width versus r_0/a at the same mesh; the $r_0/a = 0.30$ result is not mesh-converged.

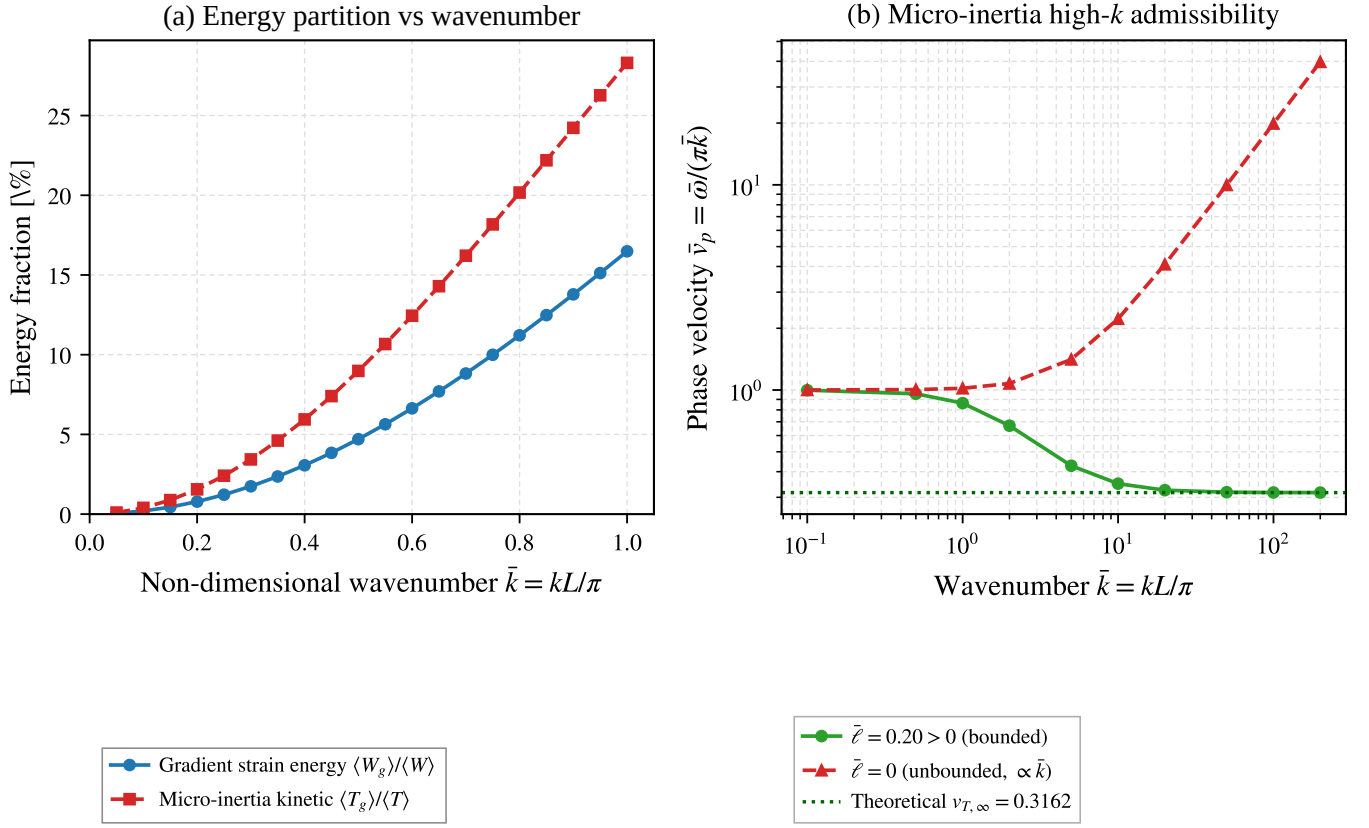


Figure 13: (a) Gradient-energy fraction W_g/W versus wavenumber. (b) Transverse-branch phase velocity with and without gradient micro-inertia; the finite high-wavenumber limit is specific to the shear branch analyzed here.

9. Conclusions

1. The conforming 32-degree-of-freedom BFS discretization applies the Bloch phase to displacement and derivative degrees of freedom alike. This provides a consistent basis for separating directional filtering from full-zone attenuation and for evaluating group-velocity steering.
2. In homogeneous Case H, aspect ratio and orientation shift the zone-edge branches and can open a stop band along selected directions, but the complete-gap measure remains negative. Orientation-based tuning therefore provides directional filtering rather than complete attenuation in this model.
3. At fixed area, varying aspect ratio redistributes the principal length scales, and the sampled directional gap responds non-monotonically. The largest sampled $\Delta_{GX} = 0.0677$ occurs at $\theta = 0^\circ$, $AR = 7$; this identifies a design point in the evaluated set, not a continuous optimum.
4. On the acoustic branch, rotating the tensor moves the steering lobes while the peak deviation remains nearly unchanged over the sampled orientations. For the evaluated ring and aspect ratios, orientation selects the energy-flow direction rather than increasing the sampled peak steering magnitude.
5. Periodic epoxy–YBCO contrast in Case C yields a positive complete gap between Bands 3 and 4 on every computed finite-element mesh, but its width narrows under refinement. The reported coarse-mesh width is therefore resolution-specific and is not a converged continuum estimate.
6. For the transverse branch studied, gradient micro-inertia gives a finite high-wavenumber phase-speed limit of 0.3162. This result is specific to the branch and constitutive model; it is not a bound established for other branches or formulations.
7. The empirical Case-H mesh rate is $p = 4.17$ over the 4^2 , 8^2 , and 16^2 meshes; the 32^2 result is resolution-limited and excluded. Seven of the eight numerical consistency checks meet their stated criteria; the energy-flux identity remains unresolved because raw harmonic fields for an independent recalculation are unavailable. The B1 comparison is visual, while B2 and B3 lack sufficient published data for external validation. These checks support numerical consistency but do not establish quantified external agreement for the two-dimensional bands.

Appendix A. High-Wavenumber Phase Velocity Asymptotics

We derive the high-wavenumber phase-speed limit for the one-dimensional transverse-shear branch of the strain-gradient model. Without micro-inertia ($\ell_i = 0$), the phase velocity grows without bound as $k \rightarrow \infty$. With micro-inertia ($\ell_i > 0$), it approaches a finite asymptote.

We consider one-dimensional transverse-shear propagation in a Mindlin Form-II gradient-elastic continuum governed by the equation of motion:

$$\mu \frac{\partial^2 u}{\partial x^2} - \frac{1}{10} \mu l_{\text{eff}}^2 \frac{\partial^4 u}{\partial x^4} = \rho \frac{\partial^2 u}{\partial t^2} - \rho \ell_i^2 \frac{\partial^4 u}{\partial x^2 \partial t^2}, \quad (\text{A.1})$$

where μ is the shear modulus, ρ is the mass density, $l_{\text{eff}}^2 = \hat{\mathbf{k}} \cdot \mathbf{L} \cdot \hat{\mathbf{k}}$ is the effective characteristic gradient-stiffness length scale (with the factor $1/10$ inherited from the second moment of the 3D ellipsoidal domain), and ℓ_i is the micro-inertia length parameter. Substituting the time-harmonic plane wave ansatz $u(x, t) = A e^{i(kx - \omega t)}$ into Eq. (A.1) yields:

$$-\mu k^2 - \frac{1}{10} \mu l_{\text{eff}}^2 k^4 = -\rho \omega^2 - \rho \ell_i^2 \omega^2 k^2. \quad (\text{A.2})$$

Solving the equation gives the dispersion relation:

$$\omega^2(k) = \frac{\mu k^2 (1 + \frac{1}{10} l_{\text{eff}}^2 k^2)}{\rho (1 + \ell_i^2 k^2)}. \quad (\text{A.3})$$

The phase velocity $v_p(k) \equiv \omega(k)/k$ is therefore:

$$v_p(k) = \sqrt{\frac{\mu}{\rho}} \sqrt{\frac{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}{1 + \ell_i^2 k^2}} = c_T \sqrt{\frac{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}{1 + \ell_i^2 k^2}}, \quad (\text{A.4})$$

where $c_T = \sqrt{\mu/\rho}$ is the classical acoustic shear wave speed.

Case 1: No micro-inertia ($\ell_i = 0$). Setting $\ell_i = 0$ in Eq. (A.4) yields:

$$v_p(k) = c_T \sqrt{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}. \quad (\text{A.5})$$

In the short-wave limit ($k \rightarrow \infty$), the phase velocity behaves asymptotically as:

$$\lim_{k \rightarrow \infty} v_p(k) = \lim_{k \rightarrow \infty} \frac{c_T l_{\text{eff}}}{\sqrt{10}} k \rightarrow \infty. \quad (\text{A.6})$$

Similarly, the group velocity $v_g(k) = d\omega/dk$ is given by:

$$v_g(k) = c_T \frac{1 + \frac{2}{10} l_{\text{eff}}^2 k^2}{\sqrt{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}} \sim \frac{2c_T l_{\text{eff}}}{\sqrt{10}} k \rightarrow \infty \quad \text{as } k \rightarrow \infty. \quad (\text{A.7})$$

Both phase and group velocities therefore grow without bound at high wavenumbers in this one-dimensional transverse-shear model when micro-inertia is omitted.

Case 2: With micro-inertia ($\ell_i > 0$). For $\ell_i > 0$, taking the limit of Eq. (A.4) as $k \rightarrow \infty$ gives:

$$v_{p,\infty} \equiv \lim_{k \rightarrow \infty} v_p(k) = c_T \lim_{k \rightarrow \infty} \sqrt{\frac{\frac{1}{10} l_{\text{eff}}^2 k^2 (1 + \frac{1}{l_{\text{eff}}^2 k^2})}{\ell_i^2 k^2 (1 + \frac{1}{\ell_i^2 k^2})}} = c_T \frac{l_{\text{eff}}}{\sqrt{10} \ell_i}. \quad (\text{A.8})$$

With velocity scaled by $c_T = 1.0$, the dimensionless limit is:

$$\bar{v}_{T,\infty} = \frac{\bar{l}_{\text{eff}}}{\sqrt{10} \bar{\ell}_i}. \quad (\text{A.9})$$

For the Case H baseline ($l_{\text{eff}} = l_{\text{iso}} = 0.20$ m, $\ell_i = 0.20$ m from Table 2, with $\bar{l}_{\text{eff}} = 0.20$ and $\bar{\ell}_i = 0.20$), the transverse asymptotic limit is:

$$\bar{v}_{T,\infty} = \frac{0.20}{\sqrt{10} \times 0.20} = \frac{1}{\sqrt{10}} \approx 0.3162. \quad (\text{A.10})$$

As verified numerically in Section 7, the finite element phase velocity at $\bar{k} = 200$ attains $\bar{v}_p = 0.3163$, matching the analytical horizon within 0.03%. For this transverse branch, micro-inertia gives a finite high-wavenumber phase-speed limit.

Appendix B. Time-Averaged Energy Flux in Form-II Gradient Elasticity

In this appendix, we derive the conservation of elastodynamic energy and the explicit form of the time-averaged Poynting-type energy flux vector $\mathbf{S}$ in a Mindlin Form-II gradient-elastic solid with micro-inertia.

The local balance of mechanical energy is expressed by the continuity relation:

$$\frac{\partial}{\partial t}(W + T) + \nabla \cdot \mathbf{S} = 0, \quad (\text{B.1})$$

where W and T are the strain and kinetic energy densities defined in Eqs. (10) and (13). Differentiating W and T with respect to time gives:

$$\frac{\partial W}{\partial t} = \sigma_{ij} \dot{\epsilon}_{ij} + \tau_{ijk} \dot{\eta}_{ijk} = \sigma_{ij} \dot{u}_{i,j} + \tau_{ijk} \dot{u}_{i,jk}, \quad (\text{B.2})$$

$$\frac{\partial T}{\partial t} = \rho \dot{u}_i \ddot{u}_i + \rho \ell_i^2 \dot{u}_{i,j} \ddot{u}_{i,j}. \quad (\text{B.3})$$

Expanding the spatial derivative terms in Eq. (B.2) using the product rule:

$$\sigma_{ij} \dot{u}_{i,j} = (\sigma_{ij} \dot{u}_i)_{,j} - \sigma_{ij,j} \dot{u}_i, \quad (\text{B.4})$$

$$\begin{aligned} \tau_{ijk} \dot{u}_{i,jk} &= (\tau_{ijk} \dot{u}_{i,j})_{,k} - \tau_{ijk,k} \dot{u}_{i,j} \\ &= (\tau_{ikj} \dot{u}_{i,k})_{,j} - (\tau_{ijk,k} \dot{u}_i)_{,j} + \tau_{ijk,kj} \dot{u}_i. \end{aligned} \quad (\text{B.5})$$

Similarly, the micro-inertial rate term in Eq. (B.3) expands to:

$$\rho \ell_i^2 \dot{u}_{i,j} \ddot{u}_{i,j} = (\rho \ell_i^2 \dot{u}_i \ddot{u}_{i,j})_{,j} - \rho \ell_i^2 \dot{u}_i \ddot{u}_{i,jj}. \quad (\text{B.6})$$

Adding Eqs. (B.2) and (B.3) and substituting the equation of motion $\sigma_{ij,j} - \tau_{ijk,k} = \rho(\ddot{u}_i - \ell_1^2 \ddot{u}_{i,jj})$ from Eq. (14) cancels the bulk terms:

$$\frac{\partial(W+T)}{\partial t} = [(\sigma_{ij} - \tau_{ijk,k})\dot{u}_i + \tau_{ikj}\dot{u}_{i,k} + \rho\ell_1^2 \dot{u}_i \ddot{u}_{i,j}]_{,j}. \quad (\text{B.7})$$

Comparing Eq. (B.7) with Eq. (B.1) gives the instantaneous energy flux vector S_j :

$$S_j = -(\sigma_{ij} - \tau_{ijk,k})\dot{u}_i - \tau_{ikj}\dot{u}_{i,k} - \rho\ell_1^2 \dot{u}_i \ddot{u}_{i,j}. \quad (\text{B.8})$$

The terms in Eq. (B.8) describe classical traction power modified by the double-stress divergence, double-stress power acting on velocity gradients, and a micro-inertial term.

For time-harmonic fields represented in complex notation $\mathbf{u}(\mathbf{x}, t) = \text{Re}[\tilde{\mathbf{u}}(\mathbf{x})e^{-i\omega t}]$, the cycle-averaged energy flux $\langle S_j \rangle = \frac{1}{T} \int_0^T S_j dt$ is given by:

$$\langle S_j \rangle = -\frac{1}{2} \text{Re} [(\tilde{\sigma}_{ij} - \tilde{\tau}_{ijk,k})(-i\omega \tilde{u}_i)^* + \tilde{\tau}_{ikj}(-i\omega \tilde{u}_{i,k})^* + \rho\ell_1^2 (-i\omega \tilde{u}_i)(-\omega^2 \tilde{u}_{i,j})^*]. \quad (\text{B.9})$$

Integrating over the unit cell Ω_{cell} , the cycle-averaged energy flux $\langle \mathbf{S} \rangle$ and stored energy satisfy:

$$\mathbf{v}_g = \frac{\langle \mathbf{S} \rangle}{\langle W \rangle + \langle T \rangle}, \quad (\text{B.10})$$

This identity relates the group velocity $\mathbf{v}_g = \nabla_{\mathbf{k}}\omega$ to the cell-averaged energy flux and stored energy. The earlier consistency table lists a group-velocity/energy-flux residual of 6.91×10^{-10} (Table 1). Because the harmonic micro-inertial sign in Eq. (B.9) has been corrected, this residual remains a prior result pending rerun, not a current verification.

Data availability statement

Numerical results are reported in the article and Supplementary Material. The raw solver-output files used to generate the plotted curves are not included in this package.

Conflict of interest disclosure statement

No potential conflict of interest was reported by the authors.

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